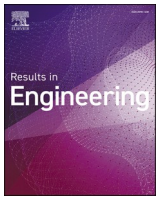
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Review article



Ultra-wideband (UWB) antennas for breast cancer detection with microwave imaging: A review

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ABSTRACT

In recent years, microwave imaging has attracted much attention and offers several advantages over existing imaging systems. It allows for a thorough examination of biological tissues, making it possible to identify changes in their shape and pinpoint their exact locations. One exciting development in this field is ultra-wideband (UWB) microwave imaging, which delivers superior results while using radiation that is not harmful. Antennas are a crucial part of UWB systems and require careful optimization, especially considering their proximity to the human body. Various factors need to be considered when designing antennas, including the choice of materials and dimensions, operational bandwidth, the impact of the human body on antenna performance, and the dynamics of short-pulse propagation. Recent research has focused on improving electromagnetic sensors used in these systems, whether as standalone units or as part of antenna arrays. This paper aims to comprehensively review significant advancements in high-performance UWB antenna sensors used in microwave imaging systems utilized for breast cancer detection.

1. Introduction

As a unique imaging technique, microwave imaging systems rely on antennas that operate across a wide range of frequencies called UWB antennas. These antennas are crucial in providing UWB cancer imaging systems with advantages, such as high image resolution, tissue penetration, and the ability to differentiate between different types of tissues [1,2]. This paper aims to provide a comprehensive tutorial that explores the principles, characteristics, and applications of UWB antennas in microwave imaging systems specifically designed for cancer detection. Microwave imaging systems use safe and non-ionizing electromagnetic waves in the microwave frequency range to create detailed images of biological tissues [3,4]. These systems provide a non-invasive alternative to traditional imaging methods like X-rays or computed tomography (CT) scans, as reviewed vastly in [5]. Despite all the advantages emphasized in these methods, there were still some drawbacks, such as the quality and accuracy of detection.

Breast cancer imaging methods may be utilized at several phases of cancer care, including detection, diagnosis, and follow-up. Mammography [6], ultrasonography, magnetic resonance imaging (MRI) [7], and positron emission tomography (PET) [8] scans are all examples of frequent imaging procedures. Because of its ability to identify anomalies, including tumors and microcalcifications, mammography is frequently utilized as a screening tool for breast cancer [9]. Screening mammography raises concerns about possible risks due to repeated exposure to ionizing radiation, and it may have limited sensitivity in women with dense breast tissue [10]. Furthermore, ultrasound is risk-free since it causes no tissue damage and uses no ionizing radiation. It can help distinguish between solid masses and fluid-filled cysts and helps guide biopsies [11]. Because it relies on the operator's skill, ultrasound could not be as informative as other methods like magnetic resonance imaging. Small lesions may also be missing.

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The thick breast tissue, multi-focal malignancies, and disease progression may all be examined using Magnetic resonance imaging (MRI). The MRI takes longer and costs more than other imaging methods. Also, it has the potential of false positives, which might lead to unneeded biopsies or other treatments.

By detecting aberrant metabolic activity, PET scans aid in the detection of metastatic breast cancer. Due to their low resolution, PET scans are often combined with other imaging techniques. Radioactive tracers are also used in these procedures.

Non-ionizing microwaves in imaging are safer than ionizing radiation methods like mammography. Microwaves easily penetrate the tissues and detect tissue electrical variations. Distinct cancerous tissue characteristics might enable differentiation. Microwave imaging offers enhanced contrast over traditional methods, aiding breast cancer identification, especially for dense tissue. Creating practical microwave imaging faces challenges such as interference, scattering, and image reconstruction. Still experimental, microwave imaging lacks the clinical testing of established methods like mammography and MRI [12].

There are various possible benefits of microwave imaging over conventional approaches when it comes to diagnosing breast cancer.

1. It detects breast cancer in patients with thick breasts more effectively than mammography.
2. Comparing benign and malignant tumors is more accurate than ultrasonography.
3. It saves money and time compared to magnetic resonance imaging.
4. Patients are not put at risk of ionized radiation exposure.

By transmitting microwave signals into the body and measuring the reflected or scattered waves, microwave imaging systems can provide valuable information about tissue composition, structure, and abnormalities like cancerous tumors [13]. UWB antennas play a critical role in these systems by facilitating the transmission and reception of microwave signals within a large bandwidth [14–17]. UWB antennas cover an extensive frequency range, from hundreds of megahertz to several gigahertz, making them apt for cancer detection in microwave imaging [18,19]. Their bandwidth, often over 500 MHz up to a few GHz, facilitates high-resolution imaging via short pulses in most imaging systems using time domain analysis [20]. This broad frequency provides the benefit of both deep penetration and high-resolution images. Additionally, UWB antennas can be compact at microwave frequencies, offering comfort and easy placement.

Careful design considerations can ensure better impedance matching for an antenna across a wide frequency range. A high-performance UWB antenna must be obtained to obtain all these advantages [21]. UWB antennas used in breast cancer imaging systems comply with strict electromagnetic safety regulations to protect the well-being of patients. The applied systems usually have at least a Tx and a Rx. These antennas can act multi-static and monostatic to affect the imaging process. For instance, the imaging system in [22] used an imaging system, but only a narrow band was used for the imaging processes, which can affect the image quality dramatically. Flexibility is also essential in wearable and imaging systems.

Therefore, a four-port multiple-input-multiple-output (MIMO) flexible antenna was designed for Ultrawideband (UWB) application using a Polydimethylsiloxanes substrate. It had dimensions of 56×41 mm^2, BW of 3.4–11.8 GHz, and 4.3 dBi peak gain. Simulation and measurement findings showed that the antenna performed well under bending situations with different curvature radius [23]. Their crucial role in achieving high-resolution imaging and enhancing breast cancer detection continues to advance through ongoing research and technological progress. For example, a special structure was constituted with patch geometry on a planar surface. The radiator consists of three counter U-shaped conducting strips united with a patch structure where two are positioned nearby, and the third geometry covers the remaining conducting strips. This corrugated design was created on a ground plane in which zigzag pattern slots are engineered to receive more than 20 dB of high isolation. The configured antenna radiates a broad frequency spectrum of 3.10 GHz–11.85 GHz. Apart from its good performance as a UWB antenna, it was assessed for imaging applications, but based on its performance, it could be used for breast cancer detection [24]. Hence, applying UWB antennas in this context will present multiple challenges and difficulties. In addition, the UWB systems face many difficulties, such as designing an appropriate antenna for a particular bandwidth and lowering antenna efficiency. Also, the message transmission speed and data rate of the UWB antenna are necessary for better communication [25].

First, there is the matter of bandwidth (BW) since UWB antennas must operate across a broad frequency range to capture the essential information for breast cancer detection. Designing antennas with wide bandwidth necessitates meticulous optimization to achieve impedance matching and desired radiation characteristics. Achieving this while preserving the desired radiation performance calls for advanced design techniques such as choosing the right antenna type, using multiple radiating elements, optimizing antenna dimensions, substrate selection, ground plane design, and more. Moreover, antenna efficiency is vital in maximizing the signal-to-noise ratio and overall system sensitivity. UWB antennas face efficiency challenges due to their wideband operation, resulting in decreased power emitted into the tissue and lower system performance [26–28]. Electromagnetic interference (EMI) is also a concern since UWB antennas emit electromagnetic energy over a wide frequency range, potentially causing interference with nearby electronic devices or systems. Effectively managing EMI and ensuring regulation compliance can be particularly demanding, especially in crowded or sensitive environments like hospitals [29,30].

Calibration and signal processing pose further obstacles, as UWB-based microwave imaging systems require precise calibration and sophisticated algorithms to reconstruct high-quality images. The intricate nature of UWB signals and the occurrence of multipath propagation in breast tissue introduce challenges in calibrating the system and developing practical imaging algorithms. Lastly, the heterogeneous nature of breast tissue presents a challenge as it exhibits variations in dielectric properties, such as permittivity and conductivity, which can affect UWB antenna performance. These variations in tissue composition and structures can distort received signals, making detecting and localizing cancerous tissue difficult. In addition, the tissues are very lossy, which requires high SNR to detect something in the imaging area (high SNR needs high gain difficulty when having a wide BW like what we get in UWB antennas) [31,32]. By deep consideration and resolving the challenges above, utilizing a proper UWB antenna with the required and outstanding characteristics, practical and feasible microwave imaging systems with high accuracy are achievable.

Several UWB antennas were designed in planar or printed form, which suits conformal systems and flexible arrays. Some others tended to radiate omnidirectionally. A four-port dual-band notch ultra-wideband (UWB) multiple input and multiple output (MIMO) antenna for automotive applications was presented. It was designed and optimized by embedding a rectangular stub onto the radiator and lowering the ground plane to realize the UWB frequency spectrum. The dual-band frequency notch, from 4.6 to 5.9 and 6.8–8.8 GHz, is achieved using a half wavelength slot onto the feedline. The obtained gain was more than 6.1 dB, which had both bi- and omnidirectional radiation [33].

Furthermore, the imaging system in [34] utilized an omnidirectional UWB antenna. Despite having good performances, these omnidirectional UWB antennas could not satisfy the UWB antennas entirely since they radiated towards the other part of the body and were not the target's area. This problem could create radiation problems on body parts and degrade imaging performance since it causes a strong coupling between the antenna and the body. Therefore, some, like [35], designed a directional UWB antenna for more focus and contrast on the imaging's desired area, ensuring broad signal coverage. However, the authors in [35] failed to test the flexibility. The working BW was insufficient since the unhealthy cells have a higher water percentage and larger dielectric

constant than healthy cells. These cells and insufficient BW also caused the difference in the back scattered signal of the antenna, which is analyzed to diagnose the tumor or cancer.

Printed monopole antennas like bowtie antennas are famous for UWB imaging applications. Printed monopole antennas, consisting of a conducting element on a dielectric substrate, offer a compact size, low profile, ease of fabrication, broad bandwidth, and acceptable radiation characteristics such as low cross-polarization and linear polarization, which are very helpful for imaging applications. This antenna offered only half of the working BW for a regular UWB antenna. Therefore, the reconstructed image of the target in the imaging environment is covered by many cluttered artifacts [36]. A compact-sized ultra-wideband (UWB) printed monopole antenna with improved gain and efficiency is presented in this article. The designed UWB antenna is fabricated on a low-cost FR-4 substrate with a relative permittivity of 4.3, loss tangent of 0.025, and a standard height of 1.6 mm, sized at 25 mm × 22 mm × 1.6 mm, suitable for wireless communication systems. The designed UWB antenna works with maximum gain (peak gain of 4.1 dB) across the whole UWB spectrum of 3–11 GHz. The antenna had the acceptable BW required for imaging purposes. However, they did not assess the antenna for an imaging application [37]. A Bowtie antenna resembling two triangles joined at the apex should exhibit a wide bandwidth, high gain, and low dispersion.; but the antenna in [38] failed to achieve these characteristics. Hence, it could not be a good candidate for imaging applications since it did not have wide BW.

A horn antenna is another type of UWB antenna and imaging application. Horn antennas were widely used in microwave imaging systems due to their wide bandwidth and high directive gain radiation pattern. The horn antenna presented in [39] used a meta surface technique to enhance the gain of a horn antenna. The horn antenna obtained a very high directive gain of almost 24 dBi. However, the BW was too narrow, and the antenna was significant, which could be problematic for real-world imaging applications. Horn antennas have a flared waveguide structure that facilitates efficient signal transmission and reception, making them suitable for high-resolution imaging and accurate localization of tumor boundaries. Considering these requirements, a substrate-integrated waveguide (SIW) H-plane filtering horn antenna with embedded metalized via-hole arrays was proposed in [40]. Two metal strips were printed on both sides of the substrate near the radiating aperture of the horn antenna to improve the BW. Apart from the enhanced BW, it is yet to be enough for imaging purposes, and the limited BW has degraded the image quality. These two horn antennas offered high performance but with large dimensions, which again reduced the scattering signal, degrading the image quality.

Another type of UWB antenna is a log-periodic antenna, which shows wide BW, is easy to integrate and has two radiation patterns. For instance, [41] provided stable impedance and radiation characteristics over a wide frequency range; however, no imaging assessment and imaging requirements were performed, and the antenna was quite large and could not obtain low gain. Another log periodic antenna design for UWB applications obtained high gain and wide BW of 26–44 GHz. It could be considered for microwave skin imaging and volume target since its working BW is high but unreliable for breast imaging [42]. Like the previous work, [43] also designed a UWB antenna for the THz band that showed good directivity, making them well-suited for UWB microwave imaging systems. A wide band log periodic antenna intended for imaging of the breast. The performance of the antenna was enhanced by using metamaterial structures. Having its BW improved, and assessment for imaging breast could not detect the tumor without significant clutter and artifacts in the imaging environment [44]. Table 1 presents some recent planar antennas having wide band and UWB for imaging applications.

Several review papers have examined the applications of UWB antennas in microwave imaging for cancer detection. For example, [45] explored wearable and non-wearable UWB antenna designs, focusing on their substrate properties and performance within WBANs. It highlighted challenges such as human-body interference, time-varying electromagnetic wave dispersion due to movement, and exposure risks. The paper also discussed design improvements and future trends in ON—OFF body antennas. However, the paper should explore a broader range of UWB antenna types to cover emerging innovations. They also failed to include a detailed comparison of various designs regarding performance metrics like Specific Absorption Rate (SAR), gain, and efficiency, which would strengthen its impact. It lacked extensive quantitative comparisons or meta-analyses, which could provide a more robust foundation for conclusions. In addition, it focused more on general principles rather than specific use cases or scenarios for WBAN applications, and it touched lightly on practical challenges like mass production, regulatory compliance, and long-term reliability, which are critical for real-world applications.

[46] offered a comprehensive exploration of UWB antennas tailored for biomedical imaging, especially in microwave imaging applications. It classified antenna designs based on manufacturing technologies, radiation mechanisms, and methods to enhance bandwidth and reduce size. The paper focused on critical performance parameters like VSWR, reflection coefficients, fidelity factor, and group delay, all essential for effective antenna performance in biomedical settings. While it covered many antenna designs, the analysis of individual designs lacked depth, which could leave readers needing more specific insights into cutting-edge designs or emerging trends. The paper did not fully explore advancements in materials, such as flexible or wearable antennas, which were becoming increasingly relevant in biomedical applications. The discussion did not extensively cover the practical challenges of integrating these antennas into clinical systems, such as compatibility with medical devices or regulatory hurdles. Although it reviewed many designs, a comparative performance table or summary of trade-offs between different designs could improve usability for engineers.

[47] offered an overview of microwave imaging (MI) techniques for breast cancer detection, discussing advancements, challenges, and comparative benefits of MI over conventional imaging methods like mammography, ultrasound, MRI, and CT. While the paper examined the advantages of MI, it did not delve deeply into the challenges of clinical

Table 1
Recent UWB antennas were designed in planar or printed form.

References	Antenna type	BW (GHz)	Peak Gain (dBi)	Efficiency (%)	Applications	Size (mm^3)
[33]	3D slots	4.6–5.9, 6.8–8.8	<2.5	N/A	WiFi	22 × 10 × 25
[34]	Omnidirectional antenna	0.51- 6.1	N/A	N/A	Breast cancer detection	N/A
[35]	Bowtie antenna	1–3	N/A	N/A	Breast cancer detection	60 × 60 × 2
[36]	Bowtie antenna	1–6	N/A	>80	Medical Microwave Imaging	60 × 60 × 50
[37]	Jug-Shaped CPW patch	3–11	4.1	<80	Wireless Communications Networks	25 × 22 × 1.6
[38]	Bowtie antenna	3.1–5	7.5	82.6	Sub-6 GHz 5G	32 × 3233.8
[39]	Horn antenna	11–14	20.3	N/A	terrestrial and space	144 × 108
[40]	SIW horn antenna	15–18	5.1	N/A	N/A	57.8 × 34 × 1.524
[41]	Log-periodic	470–860	<8	N/A	MWI	30.6 × 29.5 × 2
[42]	Log-periodic	26–44	14.29	<97	5 G communication	20 × 40 × 0.508
[43]	Log-periodic	0.09–30.14 THz	16.02	98.2	video rate, biomedical imaging	0.45 × 0.5
[44]	CPW Log-periodic	2–5	5.5	N/A	N/A	50 × 40 × 1.6

translation, such as regulatory barriers, cost, and patient accessibility. Besides, the review did not fully address the technical hurdles in MI, such as the need for advanced algorithms and high-sensitivity hardware for reliable imaging results. Furthermore, the paper primarily offered qualitative descriptions of MI techniques without providing performance metrics or quantitative comparisons, which would strengthen its technical rigor.

In contrast, our review paper distinguishes itself by addressing the limitations of previous studies by integrating the latest research published over the past five years. It offers a comprehensive examination of UWB antenna designs alongside an in-depth exploration of breast cancer diagnostic methodologies, focusing on the role of RF technology in medical imaging. Unlike prior reviews, our work emphasizes critical design parameters for UWB antennas on solid and flexible substrates, categorizing antenna types specifically tailored for breast imaging applications. Moreover, we highlight the significance of clinical trials and propose future research directions, providing a practical and forward-thinking perspective that bridges technological advancements with real-world clinical implementation. In addition, our discussion of future research directions and clinical relevance offers actionable insights for researchers and practitioners, ensuring the paper's practical utility.

This review article is configured in seven sections. An extensive introduction was given in section one, considering all the challenges of tumor detection in breast and antenna design procedures and the types of UWB antennas used for imaging. Breast cancer diagnosis techniques and radio frequency as a promising tool for cancer detection are presented in section two. The UWB antenna challenges and design parameters in breast cancer imaging are given in section three. Afterward, UWB antennas in microwave imaging of breast cancer designed on flexible and wearable substrates are discussed in section four, followed by some clinical trials in Section 5. A discussion of the possible suggestions for UWB antennas and future directions is illustrated in section six, and the paper concludes in section seven.

2. Breast cancer diagnosis techniques and integration of RF technology

Modern technology and lifestyles cause several harmful health issues; cancer is the most destructive. Many approaches are used to mitigate this danger, and the most effective strategies are screening and early detection. Nearly 25 percent of women are affected by breast cancer, and more than 2 million cases were reported globally in 2018. If detected in its earliest stage, 90 % of these patients would likely be cured within five years [48]. Existing imaging includes Ultrasound (US), Position Emission Tomography (PET), X-ray mammography, and Magnetic Resource Imaging (MRI).

Mammography (X-ray) uses low energy to construct an image of the breast. Although it initially showed promising outcomes, drawbacks include a high risk of radiation exposure for both patient and technician, which limits viability. Younger women (40 s) are more often affected but often have false negative or false positive assessments [49]. Ultrasound uses lower-frequency sound waves (~20 kHz) to visualize internal organs where the transducers transmit sound waves for which there is no known radiation risk [50]. However, the US cannot differentiate healthy from pathologic tissues [51]. Another method uses radio waves and a magnetic field for image reconstruction (MRI), often used for preoperative evaluations [52]. However, it is costly and incapable of providing early diagnostic images confirming suspect lesions. Positron Emission tomography (PET) and computerized tomography (CT) can detect early cancerous lesions, but the resulting images have low resolution [53,54]. Table 2 presents the advantages and disadvantages of breast cancer imaging techniques.

Researchers were highly motivated to develop new approaches to breast cancer imaging [19]. Recent high-performance tools with better outcomes, like radio frequency (RF) technology, were considered to cover the challenges of the previous techniques. Microwave (MW) and ultra-wideband (UWB) technologies are considered to be part of RF technologies [45]. Both emergent methods showed promising outcomes regarding processing time, high resolution, simplicity, ease of use, and costs. RF technology is commonly used in various medical applications, including cancer diagnosis and treatment. As a cancer treatment tool, RF primarily involves radiofrequency ablation (RFA) or thermotherapy (RF-TT). RFA is a minimally invasive procedure that uses heat to destroy cancer cells.

Table 2
The applied techniques for breast cancer imaging.

Techniques	Advantages	Disadvantages
Ultrasound [51]	Low frequency; no exposure to dangerous radiation; can image internal organs	Unable to differentiate healthy from unhealthy tissue; low-resolution
MRI [52]	Creates an image of organs; presurgical evaluations	Costly, cannot image early lesions; cannot be used for people with implants; low accuracy
PET [53]	Early-stage imaging	Low resolution
X-ray [55]	Minimally invasive structural imaging. 3D image reconstruction	Long process; dependent on sample size; X-ray detectors are placed far away, have low image quality, and damage human tissue.
MW/UWB imaging [56, 57]	Non-invasive; detects hidden structures; quick; differentiates tumor from healthy tissues; does not damage tissue; operates at high temperatures; insensitive to environmental conditions	Expensive; uses higher frequencies that necessitate calibration.

A thin needle was inserted into the tumor, and high-frequency electrical currents are passed through the needle, heating the tumor cells, increasing their temperature, and eventually killing them. RF-TT was a similar procedure; however, the electrical currents were used to heat the tumor cells to a lower temperature, which causes them to die by a process called apoptosis. Both RFA and RF-TT were used to treat localized small tumors. These techniques are often utilized as treatment options for patients who are not suitable candidates for surgery or other therapies. For example, radiofrequency ablation (RFA), microwave ablation (MWA), and cryoablation are standard treatments for hepatocellular carcinoma (HCC) tumors smaller than 5 cm. However, despite the combination of these technologies, their accuracy in detecting small tumors remains limited. In the case of prostate cancer, RF technology has been employed for heat treatment, successfully treating about one-third of the patients evaluated [58]. However, this heat treatment also resulted in several side effects for the patients undergoing the procedure [59,60]. This technique was also used to treat small tumors in organs such as the liver. While RFA is primarily a therapeutic modality, it can also provide diagnostic information. During the RFA procedure, the emitted RF energy creates a thermal field within the tissue [61]. By monitoring changes in the RF energy and temperature distribution, physicians can gain insights into the characteristics of the tissue being treated. This information can help assess the tumor's nature, evaluate the treatment response, and determine the extent of tissue ablation [61]. RF technology can also aid tumor detection and localization using microwave radiometry [62]. They exploited dielectric and thermal differences in tissues to pinpoint tumors. By analyzing RF signals from tissues, abnormal areas indicating cancer can be identified [63]. Apart from the reasonable outcomes using the RFA-based techniques, they had several limitations on the size of the tumors, the quality, and the detection of the actual location of the tumors. To improve the RFA technique's drawbacks, they were combined with other techniques like ultrasound (US) and computed tomography (CT) scans, which depend on tumor type, location, and individual factors [64–67].

Aside from their role as a diagnostic tool, microwaves have numerous applications in the medical field. One application involves

producing medical images by distinguishing between the dielectric properties of healthy and malignant tissues [68]. It is achieved through the interaction of electromagnetic waves with matter. The interaction between electromagnetic waves and matter relies on dielectric properties, which can be directly associated with various biological constituents such as water content, bone, fat, muscle, and others [69]. A substantial contrast between normal tissue and tumors can be achieved by employing microwave radar techniques in imaging systems.

Microwave imaging (MWI) systems are generally more affordable than some traditional imaging technologies (e.g., MRI), making them more accessible. Microwave-based imaging has the potential to detect small tumors [70,71], making it a valuable tool for early detection and intervention. Detecting tumors at an early stage can significantly improve patient outcomes. MWI can be suitable for frequent screening [72], enabling regular monitoring of high-risk individuals or those undergoing cancer treatment. In addition, MWI has the potential to outperform other cancer imaging modalities. However, this depends on the pivotal role of antennas in microwave systems. Antenna design and properties shape electromagnetic wave focus and scattering, directly impacting imaging capabilities. Optimizing antenna design enhances image resolution and depth for cancer diagnostics. Contrast quality between tumors and healthy tissue influences microwave system resolution and depth. Higher resolution aids differentiation, while adequate penetration ensures accurate mass detection. These aspects are crucial for practical cancer tests. It's important to note that validating this in a specific context demands more details about applications, frequencies, antenna designs, and imaging techniques. In addition, since having a wide bandwidth affects the resolution of the reconstructed image, the use of UWB antennas and antenna arrays as a perfect candidate for imaging biological tissue in early-stage cancer detection is presented [73]. Important antenna parameters and measures should be introduced to understand the performance of the UWB antennas for breast cancer detection. In the next section, a short review of these requirements is given.

3. UWB antenna challenges and design parameters in breast cancer imaging

Ultra-wideband (UWB) antennas have unique properties that make them particularly well-suited for medical imaging, including breast cancer detection. When UWB antennas interact with breast tissue, several technical aspects, such as frequency range, resolution, and penetration depth, influence their performance. More discussion of these interactions with supporting references is presented in the following sections.

3.1. Wide frequency range and dielectric property sensitivity

UWB antennas operate across a wide frequency range (typically 3.1 to 10.6 GHz), which allows them to interact with the varying dielectric properties of different breast tissues (e.g., fat, epithelial, and malignant tissues). Healthy tissues like adipose (fat) have lower dielectric constants than malignant tissues with higher water content. This difference in dielectric properties contrasts the backscattered signals that UWB antennas can exploit to distinguish between healthy and cancerous regions.

- **Dielectric contrast:** The dielectric constant of malignant tissue can be two to four times higher than normal breast tissue. This variation is critical for microwave imaging, as UWB signals reflect differently depending on the tissue's dielectric properties.

3.2. High spatial resolution and tumor detection

The short-duration, wideband pulses emitted by UWB antennas offer high spatial resolution, essential for detecting small tumors. The acceptable temporal resolution (less than one ns) helps resolve small structures within the breast, enhancing image clarity and detection accuracy. UWB systems can, therefore, identify tumors at earlier stages compared to conventional imaging systems, which rely on longer wavelengths and have lower resolution.

- **Temporal resolution:** UWB systems typically operate in time-domain analysis, where short pulses (nanoseconds in duration) provide higher image resolution. For instance, systems like the MammoWave create high-resolution images using ultra-short pulses, detecting small tumors and differentiating them from benign tissue [74].

3.3. Non-Ionizing and safe penetration depth

Unlike X-rays in mammography, UWB signals are non-ionizing, meaning they do not pose radiation risks to patients. It is particularly advantageous for repeated screenings. Additionally, UWB antennas can achieve deeper penetration into the breast tissue while maintaining a high resolution, which is particularly beneficial in patients with dense breasts where mammography has limited efficacy.

- **Penetration depth:** UWB systems can penetrate several centimeters of breast tissue, providing a comprehensive view without the compression or ionizing radiation required in traditional mammography. It is beneficial for younger women with dense breast tissue, where conventional methods may fail to detect tumors [75].

3.4. Low specific absorption rate (SAR)

A critical consideration in medical applications is the specific absorption rate (SAR), which measures the amount of radiofrequency energy absorbed by the body. UWB antennas are designed to operate with low SAR, ensuring patient safety while maintaining effective imaging capabilities. In breast cancer detection, SAR is carefully monitored to avoid excessive heating of the tissue, which could cause discomfort or damage.

- **SAR and safety:** Studies indicate that UWB systems used in breast cancer detection adhere to safety guidelines set by regulatory bodies, making them safe for patients. The energy emitted by UWB antennas is low enough to avoid adverse biological effects while still producing accurate diagnostic images [12].

3.5. Multistatic and monostatic configurations

UWB systems can operate in either multistatic or monostatic configurations. In a monostatic setup, a single antenna acts as both the transmitter and receiver, while in a multistatic system, multiple antennas work together to capture data from different angles. Multistatic systems are more effective for breast cancer detection as they provide comprehensive data from various perspectives, enhancing image quality and reducing false positives.

- **Multistatic imaging:** Multistatic UWB systems can improve image quality by capturing reflections from different angles, offering a more detailed representation of the breast tissue and better distinguishing malignant from benign lesions.

3.6. Challenges in dense breast tissue

One of the primary benefits of UWB systems is their ability to perform well in dense breast tissue, a limitation of traditional mammography. The dielectric contrast in dense tissues (which consist of more glandular than fatty tissue) is more subtle, making it difficult to detect tumors with conventional X-ray techniques. UWB antennas, on

the other hand, can better differentiate between tissue types due to their sensitivity to dielectric properties, making them more effective in such cases.

- **Dense tissue imaging:** Studies have shown that UWB microwave imaging performs better in dense breast tissue than mammography, which struggles to identify lesions in this context. UWB technology is a valuable tool for early detection, especially for younger patients.

Some regulatory and safety issues, such as ethical considerations related to using a new imaging technology like the UWB antenna system, should have been considered before the recent years when UWB antennas used for imaging applications were presented. Introducing this new imaging technology, such as an ultra-wideband (UWB) antenna system for microwave breast cancer imaging, raises various regulatory, safety, and ethical considerations. Below are some critical points for further reading.

3.7. Regulatory approval process

New imaging technology must undergo rigorous regulatory scrutiny before being widely implemented in clinical settings. For instance, the Food and Drug Administration (FDA) reviews medical devices, including imaging technologies, in the United States. This process involves Premarket Approval (PMA), in which UWB devices must demonstrate safety and efficacy through clinical trials before receiving FDA approval. These trials assess performance metrics such as sensitivity, specificity, and safety under real-world conditions. It also involves 510(k) Pathway. For some UWB devices, manufacturers may pursue the 510(k) pathway, demonstrating that the new device is substantially equivalent to a legally marketed device. This pathway requires less extensive clinical data but requires a comprehensive performance review.

3.8. Safety concerns

The safety of UWB microwave imaging systems is paramount, particularly in avoiding adverse biological effects on patients. Safety concerns include the Specific Absorption Rate (SAR) and Long-term Effects. Regulatory bodies set limits on SAR levels to ensure that the energy absorbed by the body is within safe limits. UWB devices must be tested to ensure compliance with these safety standards, as excessive microwave exposure can lead to tissue heating. While UWB technologies are non-ionizing, studies still need to evaluate any potential long-term effects on patients, particularly in those undergoing repeated screenings.

3.9. Ethical considerations

Developing and implementing new imaging technologies involves ethical considerations related to patient autonomy, informed consent, and the potential for disparities in healthcare access. For informed consent, patients must be adequately informed about the benefits, risks, and uncertainties of UWB imaging technologies. It includes understanding the limitations compared to traditional methods like mammography. Furthermore, considering access and equity, concerns arise regarding the availability of new imaging technologies to diverse populations. Ensuring equitable access to UWB imaging for all demographic groups is critical, especially given that some populations may have higher incidences of breast cancer.

3.10. Data privacy and security

With the integration of advanced imaging technologies, data privacy becomes increasingly important. The related issue can be Patient Data Security. As imaging systems may collect sensitive patient data, protecting this information from unauthorized access is essential. For instance, compliance with regulations such as the Health Insurance Portability and Accountability Act (HIPAA) in the U.S. is mandatory. The other concern is Data Usage Transparency. Patients should be informed about how their imaging data will be used, stored, and shared. This transparency is vital for maintaining trust between healthcare providers and patients.

3.11. Clinical implementation and training

Effective implementation of UWB imaging technologies requires adequate training for healthcare professionals. Which can involve training technicians and radiologists. Healthcare providers must be trained to operate UWB imaging devices and interpret the results accurately. This training is crucial to avoid misdiagnosis and ensure patient safety. Moreover, ongoing education programs are necessary to keep professionals updated on the latest advances in UWB technology and best practices for patient care.

Apart from the pre-presented challenges, issues, and irregularities, the UWB technology could still convince researchers and scientists to utilize it to enhance imaging by delivering more excellent resolution and better capabilities than conventional methods. UWB imaging systems must have high spatial resolution and sensitivity while maintaining a high signal-to-noise ratio for adequate resolution to detect tumors or anomalies in breast tissue early. Moreover, the interaction of UWB signals with breast tissue is complicated by the wide variety of tissue types and qualities, necessitating precise modeling and understanding of these interactions to improve imaging precision. UWB signals can be weakened and scattered when traveling through living matter, potentially reducing image resolution and depth, thereby highlighting the urgent need for methods to mitigate these effects [76,77].

Furthermore, significant signal reflections and dispersion due to breast tissue abnormalities and changes further complicate the distinction between normal and pathological tissue structures. Additionally, artifacts such as motion, interference, and undesired reflections can compromise UWB imaging quality and diagnostic accuracy [78]. Designing UWB imaging devices for clinical usage presents challenges in creating small, portable, and user-friendly systems, demanding antennas that seamlessly integrate with these setups. Meanwhile, handling the vast amount of data generated by microwave breast imaging and developing practical algorithms for data processing and image reconstruction presents substantial challenges [79]. Patient comfort and safety are paramount with non-invasive UWB imaging, requiring careful management of radiation emissions to prevent harm. Simultaneously, developing ultra-wideband imaging devices may be costly; maintaining affordability and accessibility is essential for many healthcare institutions and patients [80].

Since UWB cancer detection systems have the potential to detect different cancer types at the early stage and potentially provide high-resolution images with excellent penetration and minimum interference, they have attracted much interest in medical applications like breast cancer imaging. UWB Antennas, as a vital part of any UWB imaging system, is employed to transmit and receive electromagnetic signals, which interact with breast tissues and allow for the reconstruction of interior structures. Several parameters related to designing UWB antennas for breast cancer imaging should be considered before designing UWB antennas to achieve the best outcome for a UWB imaging system.

3.11.1. Frequency range

Ultra-wideband (UWB) antennas can operate across a broad range of frequencies. According to the U.S. Federal Communications Commission (FCC), UWB systems are defined by having either a fractional bandwidth (FBW) of more than 20 % or a bandwidth of at least 500 MHz. These antennas can cover frequencies ranging from hundreds of megahertz (MHz) to gigahertz (GHz). Two key terms help describe the frequency range of UWB antennas: the center frequency (Fc) and the FBW. The center frequency is simply the midpoint between the antenna's lowest

and highest frequencies. The FBW is a ratio of the total frequency range (bandwidth) to the center frequency [81]. Choosing the right frequency range is essential to balance signal penetration depth, clarity, and quality. Engineers use design techniques like fractal shapes or structures that use space efficiently to make antennas smaller while still allowing them to function across a wide range of frequencies. These designs help the antenna resonate at multiple frequencies and cover a wide range without taking up much physical space.

3.11.2. Group delay

Group delay describes how the time delay of an antenna changes with the frequency of the signal it handles. It is measured in the frequency domain, where the signal phase is shown as a function of frequency. Ideally, antennas with no distortion have a linear phase and a constant group delay in the right frequency range. Nonlinearities in group delay can indicate that the antenna is storing energy due to its resonant properties. Group delay also helps normalize the system's frequency response, which can improve signal quality (Fig. 1). In simple terms, group delay is the total time different signal parts pass through the antenna. The group delay must remain short, allowing for precise timing of the signals to get accurate results in UWB imaging. It ensures that signals arrive in sync, which is critical for creating clear images. In breast cancer detection, for instance, precise timing helps identify abnormalities in tissue. Therefore, controlling group delays in antennas is vital to improving the effectiveness of microwave imaging in detecting breast cancer [82].

3.11.3. Mutual coupling

"Mutual coupling" refers to the electromagnetic interaction between antennas when placed close to each other. This interaction can change the antennas' radiation patterns, impedance, and performance. In UWB imaging, like breast cancer detection, the effects of mutual coupling can be both helpful and harmful. It can distort how the antennas emit signals, cause reception problems, change impedance, and reduce the efficiency of antenna arrays (Fig. 2).

One way to reduce mutual coupling is by increasing the distance between antennas, which is not always possible in compact designs like those used for breast cancer imaging. The orientation and polarization of antennas also play a role—antennas with perpendicular (orthogonal) polarization are less affected by mutual coupling. Devices that isolate antennas, unique structures placed between them, and impedance-matching networks are used to minimize these effects. New methods in array design also help reduce coupling. For example, using decoupling metasurfaces improves signal isolation, reduces signal reflection, enhances radiation patterns, and broadens the bandwidth of antenna arrays [83].

3.11.4. Impedance matching

Impedance matching is an essential concept in antennas and microwave systems. It means ensuring that the impedance (resistance) of a transmission line or device is the same as that of the load (like an antenna). When the impedances match, power is transferred efficiently, with little energy lost. It ensures that the signal travels smoothly between the transmission line and the antenna and from the antenna to the environment and back.

If impedance matching is not done correctly, part of the energy can be reflected to the source, causing inefficient power transmission. It weakens the signal and can lead to poor reception, which affects the performance of communication or sensing systems.

In microwave imaging (medical imaging or radar, for example), proper impedance matching is crucial to get clear and accurate signals. If there's a mismatch, it can cause reflections and loss of signal quality, which harms image clarity. Poor matching reduces the signal-to-noise ratio (SNR), making it harder to differentiate objects in the image. On the other hand, good matching improves SNR, leading to better image resolution, contrast, and overall quality. One way to improve impedance matching is using "open-end slots" in the ground plane (part of the antenna). These slots help distribute the current evenly, making the antenna less sensitive to impedance changes and improving its stability. This setup can also help reduce the antenna size and allow it to work with more frequency bands [84]. For instance, for a UWB antenna, the reflection coefficient level at the desired BW should be less than −10 dB.

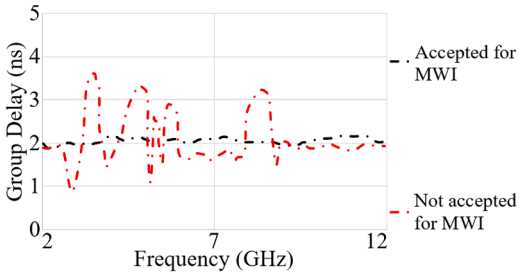


Fig. 1. A sample of accepted and not accepted MWI application.

3.11.5. Radiation pattern

An antenna's radiation pattern is a diagram showing how the energy it radiates spreads into space. It usually uses a polar or spherical coordinate system to display how vital the radiated energy is in different directions. By looking at this radiation pattern, we can understand how an antenna sends its power in various directions. In microwave imaging, the shape and characteristics of the radiation pattern are fundamental. The radiation pattern shows how sensitive the antenna is in specific directions. Designing an antenna with a stable radiation pattern is crucial for ensuring accurate and precise images. Fig. 3 (in the original context) shows what this radiation pattern looks like at the central frequency and the edges of the antenna's working range.

In this case, the radiation pattern remains steady across different frequencies, with changes of less than 20°. Antennas for sending or receiving radiofrequency signals in specific directions can focus the radiation, helping to direct microwave signals during imaging. For example, antennas with a narrow beamwidth pinpoint where a signal comes from and ensure that the radiation adequately covers the area of interest [85]. Figs. 3 and 4 can be good examples of radiation patterns.

The way antennas send out signals, known as the radiation pattern, plays a crucial role in producing clear images in imaging systems. When the radiation pattern is narrow and focused, it helps the system see small details more clearly, allowing it to better distinguish between objects that are close together. Directional antennas with narrow beams can reduce interference and improve the signal quality, leading to more explicit images. In microwave imaging, the radiation pattern of antennas affects how well they can pick up signals. While omnidirectional antennas collect signals from all directions, directional antennas, with their specially designed beam patterns, can enhance image clarity and minimize background noise [86].

In microwave imaging, the way antennas send out signals, or their radiation patterns, helps control the effects of signals bouncing off surfaces, known as multipath effects. Different types of antennas have different patterns. For example, isotropic antennas provide a uniform signal in all directions, serving as a standard reference. On the other hand, microstrip patch antennas have radiation patterns that change based on their size and how they are powered. The patterns for these antennas are often described separately in two planes: the E-plane and the H-plane. Other antennas, such as Yagi-Uda antennas [87], are directional, meaning they focus their signals on specific directions and have both primary and side lobes. Horn antennas can also create a narrow beam, although they tend to be wider at lower frequencies. For breast cancer imaging, antennas with a wide bandwidth (BW) can be more efficacious [88]. If these antennas can achieve a wide bandwidth like UWB antennas, they become more practical for this type of imaging.

By optimizing the design, it's possible to create a UWB antenna that

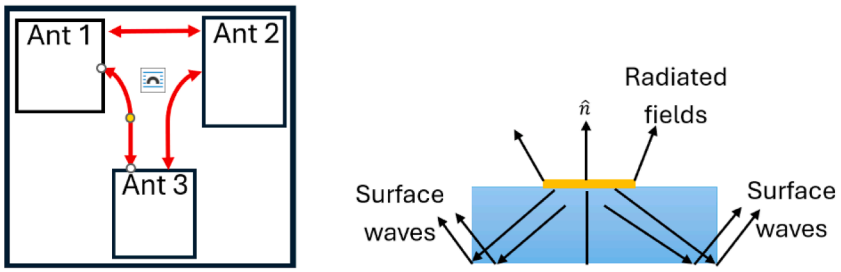


Fig. 2. A brief and visual explanation of mutual coupling.

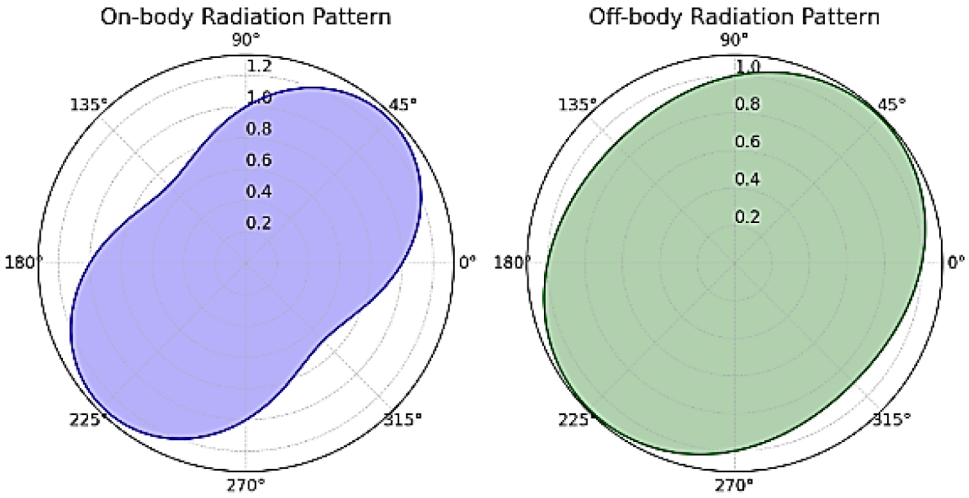


Fig. 3. Radiation pattern of an antenna for on-body and off-body scenarios in which the on-body pattern is more distorted due to interaction with the body, while the off-body pattern is more uniform.

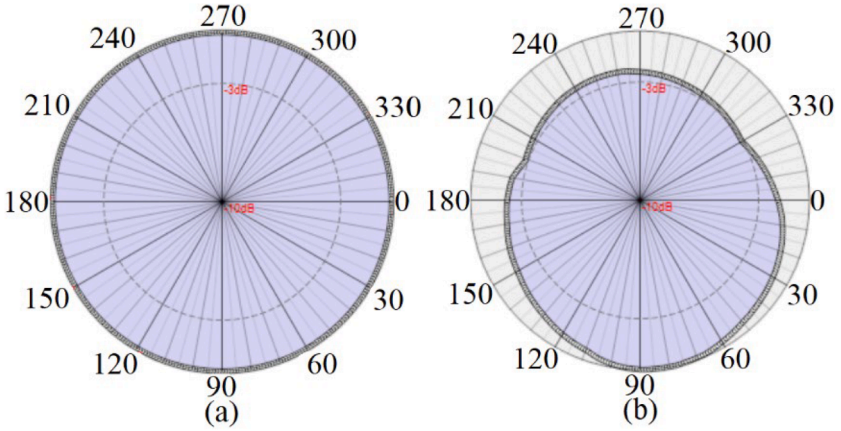


Fig. 4. The radiation patterns: (a) Omnidirectional, (b) Directional.

has consistent radiation patterns across a wide range of frequencies. It involves carefully selecting the antenna components, feeding methods and shapes to reduce variations. A well-designed radiation pattern ensures that signals can penetrate breast tissue effectively and collect clear reflections for imaging. A suitable radiation pattern improves the ability to detect abnormalities, enhances image clarity, and helps accurately identify tissue irregularities. It ultimately increases the overall effectiveness of microwave imaging.

3.11.6. Polarization

Antenna polarization refers to the direction in which the electric field of an emitted electromagnetic wave moves. There are different types of polarization, like linear, circular, and elliptical. In linear polarization, the electric field stays in a fixed direction, while in circular and elliptical polarization, the field rotates as the wave travels. How well a signal is received depends on whether the polarization of the transmitting wave and the receiving antenna match (Fig. 5). When they align, signal reception improves, enhancing image clarity in systems like microwave imaging. With the growth of wireless networks, antennas that can change their polarization (called reconfigurable antennas) have become more common. In this context, a new cut-ring microstrip antenna design has been proposed. This antenna can switch between two different polarization modes. Its simple design comprises a ring-shaped radiation patch, two PIN diode switches, and six non-metallic columns [89]. In

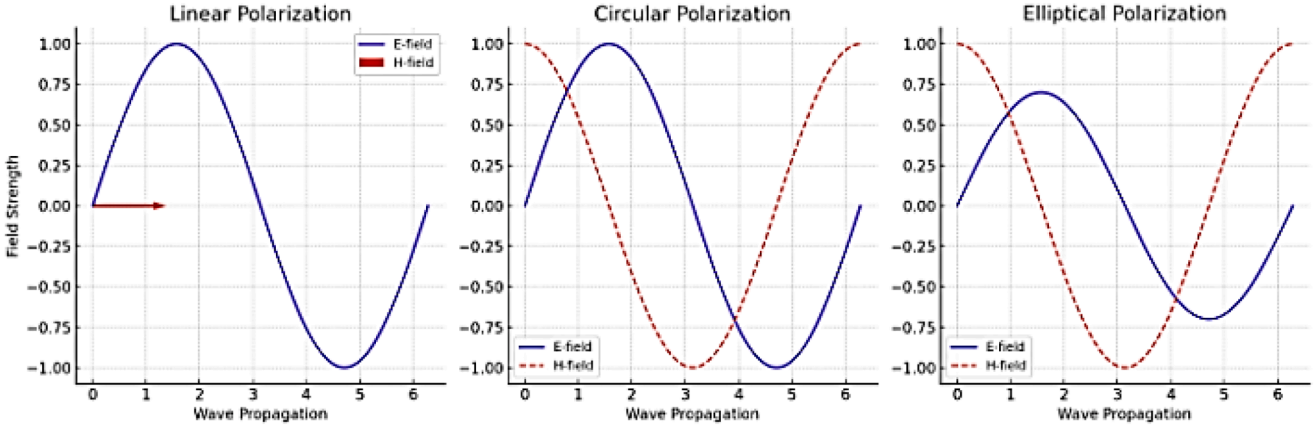


Fig. 5. Linear, circular, and elliptical polarizations of antenna representing how the electric (E) and magnetic (H) fields behave in these polarization types.

microwave imaging for breast cancer, linear polarization is often used because it helps better penetrate signals through breast tissue. When the polarization of an antenna is well-matched, it maximizes how the signal interacts with the target, allowing for better penetration and reflection through tissues. In breast cancer detection, having the correct antenna polarization improves the ability to capture detailed tissue information, which leads to more explicit images and helps identify abnormalities more accurately. This means that antenna polarization plays a crucial role in the effectiveness of microwave imaging for breast cancer detection.

3.11.7. The gain

Antenna gain tells us how well an antenna can focus its energy in a specific direction compared to a perfect antenna that sends energy equally in all directions (called an isotropic radiator). Gain is usually measured in decibels (dB) and shows how concentrated the signal is (Fig. 6). A Vivaldi antenna is an example of a high-gain antenna, meaning it can focus its energy very effectively. Vivaldi antennas can be adjusted to improve performance over frequencies [90]. For instance, adding a 3-D phase adjustment unit to a Vivaldi antenna can make it work better by increasing its gain (how much power it sends or receives) and improving how consistent its performance is across a wide range of frequencies. This technique allows the antenna to change how it handles different frequencies, enhancing its ability to focus signals across those frequencies. As a result, the antenna works better over a wide range of frequencies.

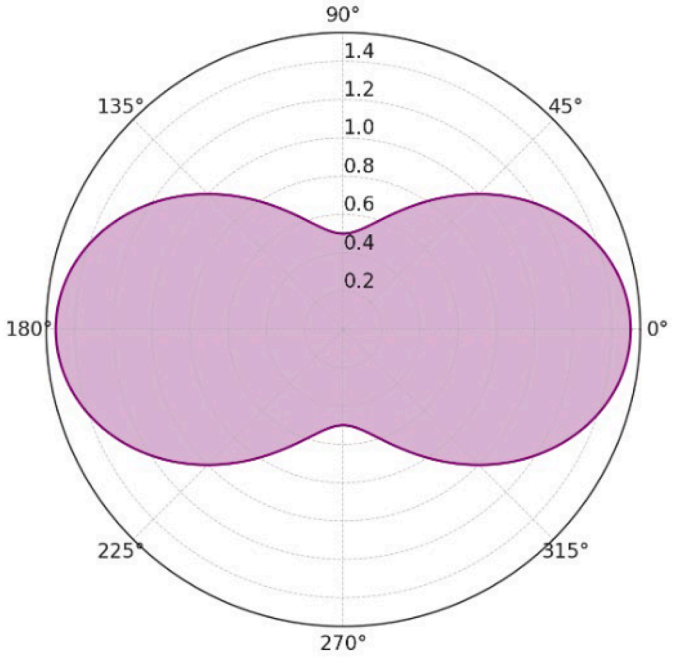


Fig. 6. A typical directional gain pattern, with higher gain in specific directions.

In microwave imaging, antennas send electromagnetic waves to scan an area, like during breast cancer detection. Antennas with higher gain can focus their energy more precisely in one direction, making the signal stronger at the target. This stronger signal improves image clarity, helps detect minor details, and increases the range at which signals can be received from farther away. Higher gain antennas also enhance image quality by sending stronger signals that can pass through tissues with less interference, making it easier to spot abnormalities, such as tumors. The gain of an antenna measures how well it directs energy and how efficiently it operates. Higher gain results in stronger signals, better tissue penetration, and more explicit images to accurately identify issues.

Regarding breast cancer detection, increasing the antenna's gain improves image clarity, sharpens details, and makes it easier to find abnormalities like tumors. This makes microwave imaging more effective for detecting breast cancer. However, high-gain antennas have some challenges. First, they can be expensive to design and build. They are also larger than other antennas, which can be an issue for smaller or compact devices. Additionally, the way the antenna sends and receives signals (its polarization) may not always be ideal, which can reduce performance in certain situations. Finally, because high-gain antennas focus energy in a narrow direction, their coverage area is limited, meaning they can only scan a minor part of the body simultaneously.

3.11.8. Size and form factor

The size and shape of an antenna are significant in microwave imaging. Size and factor depict how well the antenna can resolve details, detect signals, focus energy, and send or receive electromagnetic waves. Usually, the length of an antenna needs to relate directly to the wavelength of the frequency on which it operates. For an antenna to work well, its dimensions should be like the wavelength of the signals it handles. In microwave imaging, where the wavelengths are short, antennas must be appropriately sized to interact effectively with the waves used for imaging. A recent innovation introduced a compact antenna design with a wide range of operational frequencies, good suppression capabilities, and efficient use of space [91]. This new design is advantageous because it takes up less room and minimizes interference with other nearby devices, making it ideal for complex systems.

The size of an antenna plays a significant role in how well microwave imaging works. Smaller antennas often have lower gain, meaning they do not focus energy. While it's possible to create wideband antennas

(which can operate over a range of frequencies), they tend to be less efficient. The size of the antenna's aperture (the opening through which it emits or receives signals) affects its beamwidth, which is how narrow or wide the signal is. A larger antenna can direct signals more effectively and achieve higher gain, making it better for imaging. In microwave imaging, a focused beam is crucial because it helps signals penetrate deeper into tissues and reduces interference. However, there's a challenge: the physical space around the body is limited, requiring smaller antennas or arrays of antennas. Designing UWB antennas for cancer imaging is tricky because making them smaller often reduces their bandwidth (the range of frequencies they can use). So, engineers face the challenge of finding a balance between the size of the antenna, its bandwidth, and its gain to ensure practical imaging. After evaluating the necessary design parameters for UWB antennas, recent advancements in UWB antennas used for breast cancer imaging have been analyzed, along with their limitations and performance characteristics.

4. UWB antennas in microwave imaging of breast cancer designed on flexible and wearable substrates

UWB antennas are pivotal in microwave imaging, offering high-resolution imaging, precise localization, and effective signal penetration. Not only do they excel in medical imaging, but they also have applications in ground-penetrating radar and through-wall imaging. In microwave imaging, UWB antennas transmit nanosecond-range pulses, which scatter through targets and are captured by receiving antennas. Analyzing received signals extracts internal object details. UWB imaging antennas require wide bandwidth, directional patterns, and efficient energy transfer for optimal imaging.

A single UWB antenna or an antenna array balances range and tissue penetration. Electromagnetic waves from 1 GHz to 10 GHz efficiently penetrate tissue with moderate attenuation for imaging. Above 10 GHz, skin scatter can occur. Highly directional microwave sensors characterize tissues, aiding cancer detection. Permittivity changes are detected, high at lower frequencies due to cell membranes and low at higher frequencies due to scattering [92]. To achieve successful image reconstruction, the design of microwave imaging systems should address challenges related to antennas' characteristics, such as size, bandwidth, group delay, and transmission distortion. In addition to designing a high-performance UWB antenna, employing advanced experimental setups with breast phantoms is crucial for accurately detecting small tumors. This process involves comparing and analyzing signal changes in the presence of tumors. Specifically, it requires isolating tumor signals and back-projecting them into a model that considers the properties of surrounding tissues. Tumor locations are then identified through entropy and energy concentration analyses, enhancing the accuracy of tumor detection.

Various UWB antennas in microwave imaging systems have been employed to cater to specific application requirements, such as planar antennas like monopoles [93,94]. [93] explored a compact, cost-effective, printed monopole antenna design for biological tissue imaging applications, particularly breast cancer detection. The antenna operated across a wide frequency range (2.7 GHz to 11.4 GHz), which covered the ultra-wideband (UWB) spectrum essential for high-resolution imaging. However, the used FR4 substrate had a relatively high loss tangent ($\tan \delta = 0.02$), which might reduce efficiency and precision, especially in lossy biological environments. While compact, its dimensions still pose challenges for integrating into dense systems or requiring highly miniaturized devices. The broad bandwidth might overlap with other UWB systems, potentially leading to signal interference if not properly managed. In addition, imaging performance heavily depended on accurate modeling of tissue properties at specific frequencies; variability in these properties across patients could impact reliability. The validation used simplified breast phantoms, which did not fully replicate the complexities of human tissue, potentially affecting real-world performance. [94] designed a compact ultra-wideband (UWB) planar monopole antenna designed for 3.1 GHz–10.8 GHz applications. The design employed a ribbon-shaped slot in the radiator to enhance bandwidth performance and omnidirectional radiation patterns. While effective for UWB, the gain was insufficient for deeper penetration in dense biological tissues, limiting its utility for large or deep tumors. When used for biological tissue analysis, the FR-4 substrate had a relatively high dielectric loss, which might affect signal attenuation and imaging quality. Although it covered a wide range, the antenna might require additional tuning to focus on specific sub-bands critical for breast cancer imaging, where frequency-dependent tissue characteristics were pivotal. While promising for UWB applications, the antenna design did not explicitly address challenges such as impedance matching with human tissues, which is essential for accurate imaging results. Non-planar antennas, like horn antennas, were also used for UWB applications [95,96]. [95] designed a novel horn antenna optimized for ultra-wideband (UWB) applications in high-power microwave systems. However, while promising for general high-power and UWB applications, the paper did not focus on impedance matching or minimizing signal losses specific to human tissues. While useful for some applications, its high-power capabilities might pose risks of overheating or other adverse effects on biological tissues if not carefully managed. Although compact for general purposes, the horn structure might still be bulky for wearable or implantable systems commonly used in medical diagnostics. The antenna's performance in actual biomedical imaging scenarios (e.g., testing on phantoms or tissue models) was not discussed, leaving uncertain practical efficacy for breast cancer detection. [96] presented an innovative ultra-wideband (UWB) horn antenna fabricated using 3D printing techniques. Their nickel coating, while cost-effective, introduced non-ideal conductive effects, leading to deviations between simulated and measured performance, especially at higher frequencies. The Gain was optimized for the lower range of UWB frequencies. The radiation pattern became less directional at higher frequencies, which might reduce imaging precision in specific applications. While lightweight, the PLA material and coating were less durable compared to fully metallic designs, especially in challenging environments or repeated use scenarios. The aperture shape and size were also optimized for the specified frequency range; scaling to different frequency bands required significant redesign, potentially limiting flexibility. [97] focused on designing and applying a pentagonal fractal antenna optimized for microwave imaging, particularly for detecting abnormalities such as breast cancer. Their peak gain of 3.6 dBi was relatively low compared to other high-gain antennas, which could limit signal strength in certain scenarios. Rogers RT/Duroid 5880 increased fabrication costs compared to more widely available materials like FR-4. Besides, it was optimized for breast cancer imaging, and the antenna's design required modifications for other biomedical or industrial imaging applications. Moreover, while beneficial for bandwidth, fractal geometry could complicate the fabrication process compared to simpler monopole designs. [98] offered a novel design enhancing the performance of antipodal Vivaldi antennas (AVA) for UWB applications. The antenna operated over a wide frequency band (2.6–17 GHz), integrating a metasurface and etched notches to improve impedance matching, gain, and directivity. While having high performance, the antenna's metasurface structure could increase fabrication challenges and costs compared to simpler designs. Its extensive real-world testing, especially in biomedical imaging, was not discussed in detail, and the design might require significant alterations to adapt to narrowband systems or other non-UWB use cases. [99] designed a planar array of ultra-wideband (UWB) active slot antennas for breast microwave imaging. Operating in the 3–8 GHz range, the array integrated low-noise amplifiers directly into the slot antennas to enhance sensitivity and maintain stable radiation characteristics. This design achieves low mutual coupling and high isolation between elements, which is essential for accurate imaging in biological tissues. While showing these outcomes, it's reliance on active components increased design complexity and power consumption. Its planar configuration faced challenges adapting to irregular geometries

or breast compression setups. Besides, active elements' fabrication costs and thermal management could limit widespread deployment. Last but not least, [100] designed a compact, ultra-thin UWB microstrip antenna for microwave imaging applications. This design employed a novel configuration optimized for high performance, including features such as wide impedance bandwidth (3.1–12 GHz), consistent radiation characteristics, and omnidirectional performance in the H-plane, which were critical for imaging applications. Nevertheless, the gain at lower frequencies (e.g., 3.5 GHz) was relatively low (1.02 dB), which could limit performance in deeper tissue imaging. It exhibited omnidirectional patterns in the H-plane and quasi-omnidirectional patterns in the E-plane, which cannot be ideal for breast-focused imaging applications. The used FR4 as the substrate might not provide the optimal dielectric properties compared to advanced materials like Rogers or Taconic for certain specialized applications. These antennas were crucial in receiving or transmitting electromagnetic energy for imaging purposes. Each type has advantages and disadvantages that should be considered for breast cancer imaging. Each of them was analyzed, and their limitations for imaging applications were shown. Some recent UWB antennas' characteristics are compared in Table 3.

4.1. UWB antennas for breast cancer detection designed on flexible and wearable substrates

Wearable UWB antenna systems for medical devices, particularly for microwave imaging of breast cancer, have garnered significant interest due to their non-invasive nature, real-time imaging capabilities, and potential for early cancer detection. However, their development and design process come with unique challenges and considerations. Below is an explanation of the development conditions, viability, and design issues associated with these systems, followed by a list of relevant articles.

1. Conditions in the Development of Wearable UWB Antenna Systems
1.1. Biocompatibility and Flexibility: Wearable antennas must be designed using flexible and biocompatible materials, especially since they will directly contact the human body. These materials should ensure patient comfort and safety without compromising the antenna's performance.
1.2. Broadband Performance: UWB antennas are characterized by a large bandwidth (typically 3.1 GHz to 10.6 GHz) critical for high-resolution imaging. Maintaining a stable performance across this wide frequency range is essential.
1.3. Body Proximity Effects: Since these antennas are worn on the body, their design must account for detuning and performance degradation caused by body proximity. The human tissue, being lossy, can absorb the signal and affect antenna radiation patterns and impedance matching.
1.4. Miniaturization and Wearability: For real-life applications, these antennas need to be lightweight, low-profile, and comfortable for prolonged wear. Achieving this while maintaining good radiation efficiency and bandwidth is a challenging task.
1.5. Conformability and Durability: The antennas must maintain consistent performance even when bent or flexed with body movement. This requires the use of flexible materials like conductive textiles or stretchable polymers.
2. Viability of Wearable UWB Antenna Systems for Breast Cancer Detection
2.1. Non-Invasive and Safe: Microwave imaging is non-ionizing, unlike X-rays, making UWB antennas safer for repeated use, an essential factor in cancer screening and monitoring.
2.2. High Spatial Resolution: Due to their wide bandwidth, UWB antennas offer high spatial resolution, which is crucial for detecting small tumors early in breast cancer.
2.3. Penetration Depth: UWB systems can provide sufficient penetration depth into human tissue, which is essential for breast cancer imaging.
2.4. Cost-Effectiveness: Compared to traditional imaging techniques like MRI or mammography, UWB-based microwave imaging could be more cost-effective, making it suitable for widespread screening programs.
3. Design Issues in Wearable UWB Antennas for Microwave Imaging
3.1. Antenna-Tissue Interaction: One of the most significant design challenges is managing the antenna and tissue interaction. The human body affects the antenna's impedance and radiation pattern, so proper matching layers or isolating techniques must be used.
3.2. Polarization and Directionality: For practical imaging, the antenna should be able to transmit and receive signals efficiently with minimal interference from the surrounding environment. Designing antennas with appropriate polarization and directionality is critical to minimize the impact of multipath and signal distortion.
3.3. Impedance Matching: The antenna impedance must be well-matched with the tissue impedance over the wide UWB frequency range to maximize power transfer and minimize reflections. Achieving this consistently can be complicated due to body variability (e.g., fat, muscle, skin thickness).
3.4. Array Configuration: UWB antennas for microwave imaging are typically deployed in arrays to enhance spatial resolution. Proper design of these arrays is necessary to avoid mutual coupling and to ensure accurate imaging [101–104].

The following articles presented the most recent high-performance works to understand better the requirements and features of flexible and wearable UWB antennas.

It is worth mentioning that various flexible and wearable structures like magnetic conductors to electromagnetic designs [105] and metamaterials [106] were utilized that could enhance microwave imaging antenna performance. For instance, [105] explored the design and application of a cost-effective antenna-array integrated with metamaterials for detecting early-stage breast tumors. This system leveraged microwave imaging to differentiate between healthy and cancerous tissues by exploiting variations in dielectric properties. However, the approach was primarily evaluated in simulation or laboratory settings; its performance in real-world clinical scenarios might vary, and ensuring consistent performance of metamaterials in diverse human tissue

Table 3
Some recent UWB antennas were designed for microwave imaging.

References	Antenna type	BW (GHz)	Peak Gain (dBi)	Efficiency (%)	Applications	Size (mm^2)
[93]	DGS monopole	2.7–11.4	6.6	N/A	Biological Tissue Analysis	40 × 36
[94]	CPWG monopole	3.1- 10.8	<6	<98	Breast cancer detection	12.5 × 25
[95]	Horn antenna	2.7–12.5	8.57	N/A	high power microwave	60 × 44 × 71
[96]	Horn antenna	0.5–3.5	<6	N/A	UWB radar	78 × 37 × 45
[97]	CPW patch	3–12.7	3.6	88	Microwave Imaging	24 × 30
[98]	Vivaldi	1.5–55	N/A	N/A	5 G mm-Wave communication, microwave imaging	96 × 175
[99]	Active Slot Antenna	3–8	20	<95	Microwave imaging	77 × 81
[100]	Monopole patch antenna	3.1–12	4.4	N/A	Microwave imaging	N/A

environments showed complexity. Although metamaterials improved it, the resolution still lagged other imaging modalities like MRI. [106] developed an antenna to operate in the THz frequency range, which was sensitive to the dielectric properties of biological tissues, allowing for the detection of tumors based on their distinct electromagnetic signatures. Hence, it faced limitations when trying to image deeper tissues due to the lower penetration depth of THz waves. This THZ imaging system might require careful alignment and could be sensitive to environmental factors such as humidity and temperature, which complicate real-world applications. It also was affected by the choice of substrate.

Accordingly, [107] developed a compact, flexible antenna for ultra-wideband (UWB) applications, focusing on wearable and implant communication systems. This antenna operated effectively for on-body communications, making it suitable for healthcare applications, including monitoring systems that could aid in medical diagnostics like breast cancer detection (Fig. 7). While the dual-band operation was beneficial for specific applications, it might limit the antenna's ability to cover a broader frequency spectrum necessary for more complex UWB systems. Although using a semi-flexible substrate was advantageous, it may not be ideal for ultra-low-cost mass-market applications, as these materials can be more expensive. It also faces challenges in achieving deeper tissue penetration, critical for specific medical imaging applications such as breast cancer detection. This limited their performance in detecting tumors at deeper levels within the body.

These studies highlight the continued progress and various design strategies being employed to overcome the challenges in wearable UWB antenna development for medical imaging, specifically in breast cancer detection. Integrating these systems into wearable devices presented a promising future for non-invasive, cost-effective cancer screening technologies. For instance, a flexible UWB antenna optimized parameters for enhanced capabilities (Fig. 8). It maintained bandwidth while enabling smaller antennas suitable for breasts of various sizes. The antenna's broad frequency range is suited to biomedical applications, particularly breast cancer detection [108]. However, the antenna's efficiency was less than 70 %, which degraded its radiation and affected its image quality. Besides, the cost of more complex fabrication processes or reduced performance in certain conditions was also overlooked. The Specific Absorption Rate (SAR) safety standards must be considered in wearable antennas that use microwave frequencies as prolonged exposure may impact health, which was not addressed in [108].

It is worth mentioning that the early detection of breast cancer is vital for reducing mortality rates. Tumors that are less than 2 cm in diameter are considered to be early-stage tumors, and they have a high cure rate. In [109], a compact and flexible breast tumor sensing system was proposed. They were using dual-polarized UWB antenna arrays. This system, comprised of UWB units with circular monopoles on flexible Kapton polyimide, achieved high fidelity and stable sensing capabilities. By analyzing reflected and scattered signals from the antennas, 3D images of the breast can be reconstructed using a delay-and-sum (DAS) algorithm. The dual-polarized UWB antennas enable accurate tumor localization and sizing, with detection possible for tumors as close as 15 mm apart. The system's compact size and flexibility make it suitable for wearable applications. The developed antenna operates within the UWB frequency range, improving data capture and enhancing tumor localization and image resolution [109]. Nevertheless, the designed antenna had to effectively detect more tumors with smaller sizes. Moreover, UWB signals faced attenuation in dense tissues, which limited their effectiveness. It also posed challenges in ensuring stable signal reception on non-uniform body surfaces.

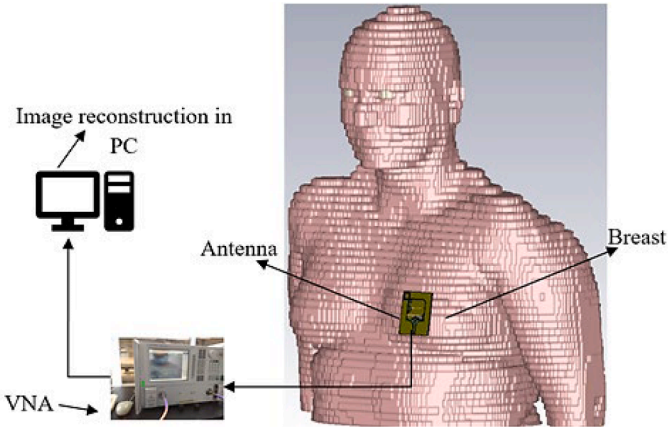


Fig. 7. Flexible on-body antennas [45].

An antenna's fidelity factor gauges its ability to transmit or receive signals accurately without distortion or degradation. Zhang and Li [110] introduced a compact design of dual-polarized flexible UWB antennas with exceptional fidelity for breast cancer detection. While flexible and dual-polarized, the antenna's coverage area might be limited depending on the body's shape and the positioning of the sensor. For accurate imaging, multiple antennas or an array setup could be needed to cover the full breast tissue. Though suitable for resolution, deep penetration into tissues with varying dielectric properties potentially limited the depth of tumor detection. While flexibility was an advantage, integrating these antennas into practical, effective, and comfortable wearable devices was still challenging. In addition, ensuring that the antenna works well across various body types and sizes has yet to be a practical consideration.

By leveraging polarization diversity, they reduced array size while maintaining tumor detection accuracy, suiting wearable breast imaging [111]. The design relied on a specific substrate material (FR-4), which might limit its application in real-world, body-integrated environments due to flexibility, biocompatibility, and potential signal attenuation in human tissues. Additionally, while the metamaterial-inspired design offered improved performance, the computational complexity involved in optimizing such an array posed challenges for practical, low-cost implementations. Furthermore, while the study presented promising theoretical results, real-world testing in clinical environments with actual breast tissue samples was necessary to validate the practical effectiveness of this design. [112] discussed using an iteratively corrected coherence factor delay and sum (ICCF-DAS) method for breast phantom imaging, a technique to enhance the quality of microwave imaging systems for detecting breast tumors. The method iteratively improved the coherence factor used in traditional delay and sum (DAS) algorithms, enhancing image clarity and tumor detection accuracy. However, the limitations included the need for high computational power and potential challenges in real-time processing, especially for clinical applications [112].

Medical imaging (MI) technology and body area networks (BANs) worked with interest in the UWB healthcare sector, focusing on and emphasizing the importance of antenna design. For instance, a dual wide-band antenna was designed for WBAN application to take care of the body's vital signs. The was privileged by a combination of rigid and wearable substrates showing high performance, like a high gain of 8.2 dBi and working bands of 1.7–3.3 GHz and 4.15–10 GHz. The antenna could be considered for imaging purposes due to its wide bands, both lower and higher bands, and it also offered compactness and comfortability for the patients. While effective in communication systems, the antenna's signal might not penetrate deeply enough into dense tissues, limiting its ability to detect tumors at deeper layers of the breast. It also required precise placement to maintain optimal signal reception and quality for imaging. Therefore, these limitations might create some clutter in the imaging area of the breast due to the stopband between the two wide bands [113].

A flexible, wearable metamaterial-based UWB antenna was designed for breast imaging and wireless body area network (WBAN) applications. The antenna, made from flexible materials, features a low-profile, planar structure. Six metamaterial unit cells with a modified grain rice

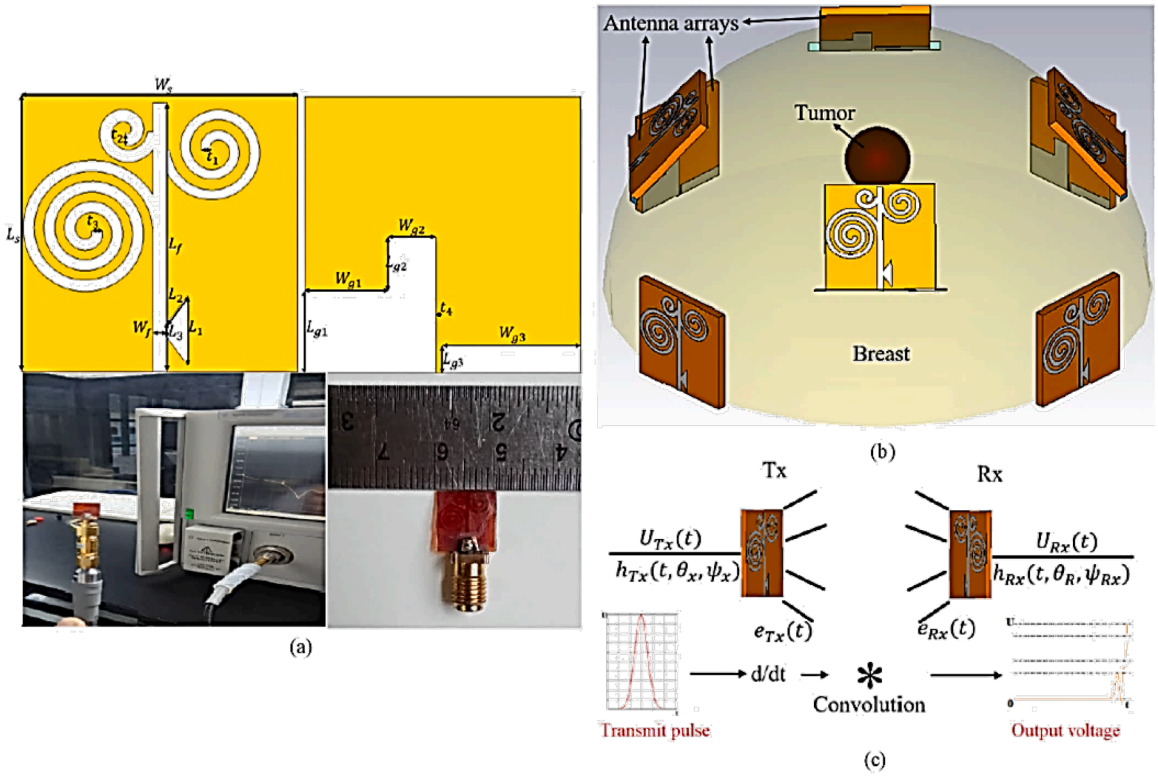


Fig. 8. Miniaturized UWB flexible antenna: (a) Simulated and fabricated prototype, (b) Simulated setup of breast, (c) Sending and receiving signals [108].

shape within a split-ring resonator were integrated into the design to improve bandwidth, gain, and directivity while reducing the specific absorption rate (SAR) to below 2 W/kg. The antenna covered a wide frequency range (6.5 GHz to 35 GHz) and achieved a maximum gain of 8.85 dBi, a directivity of 10 dBi, and a radiation efficiency exceeding 70 % across the operating band. The simulation and experimental results showed strong agreement in detecting a tumor as small as 4 mm in diameter, regardless of its position within the breast. The paper offered fascinating outcomes for imaging of the breast. While flexible, its performance might vary across different body types and skin conditions, affecting consistency in breast imaging. Incorporating the antenna into practical wearable systems for medical applications was challenging and required careful consideration of user comfort and reliability during extended use, which was considered in their assessment. Furthermore, the flexibility assessment and experimental evaluation of antenna characteristics on the breast were not evaluated [114]. Two distinct vests were designed and tested on realistic 3D human tissue phantoms representing different breast density types. The measured outcomes were validated through simulations conducted on anatomically accurate voxel models for electromagnetic analysis. The radio channel characteristics of tumor and reference cases were examined and compared using antennas embedded within the vest. Both experimental measurements and simulations demonstrated that the vest could detect tumors as small as 1 cm in diameter. Moreover, the simulation results indicated the potential to detect tumors as small as 0.5 cm. While the vest was flexible, practical issues related to the fit and comfort of the device in real-world scenarios, such as its wearability for extended periods, remained unclear. The system's effectiveness at detecting tumors located deeper within the breast tissue or at certain tissue interfaces is impacted by the limitations of microwave signal penetration, particularly in denser tissue types. Moreover, as a proof-of-concept study, there was limited clinical validation with actual patients, which means the system's reliability in real-world clinical settings and its ability to detect a wider variety of tumor types (e.g., different morphologies) was yet to be investigated [115]. A different study proposed integrating a communication system into a garment that could be used for early detection of breast cancer by utilizing a UWB wearable antenna. Early detection is crucial and has long-term benefits for women. The proposed UWB antenna operates within a frequency range of 3.1–10.6 GHz, maintaining a reflection coefficient of −10 dB. The fabricated antenna used conductive and non-conductive wearable materials, including Shieldit Super and felt. The performance of the UWB antenna in wet environments was analyzed through four stages: submerged in water, immediately wet, nearly dry, and completely dry. One limitation was the potential complexity of real-world application, particularly regarding the effect of human body movements and the variability of skin conditions, which could influence the antenna's performance in clinical settings. Additionally, while the study addressed wet performance, its results were based on specific moisture stages (e.g., entirely dry or wet), which might not fully reflect dynamic environmental conditions encountered during regular use. In addition, the pattern was also omnidirectional, which affected the antenna characteristics [116]. An antenna element operated over a bandwidth of 2.4 GHz to 10.6 GHz when using an FR4 substrate, which extended to 2.57 GHz to 12.6 GHz when Dacron fabric was used instead. In the proposed detection system, one antenna was dedicated to transmission while others were used to receive the transmitted signal. When Dacron fabric served as the substrate, the antenna achieved gains of 5.49 dBi, 9.87 dBi, 11.9 dBi, and 14.7 dBi at resonant frequencies of 2.84 GHz, 3.87 GHz, 5.83 GHz, and 8.24 GHz, respectively. The antenna exhibited a flexible design with minimal vertical and horizontal bending impact throughout the operating frequency range. The system in [117] was evaluated using heterogeneous breast phantoms, which, while useful for simulation, did not fully replicate the complexities of real human tissue. This limited the ability to assess how the antenna array would perform on patients with varying breast tissue densities and other anatomical factors. Although the antenna performed well across the 2.4 GHz to 12.6 GHz frequency range, the practical effectiveness of UWB antennas in detecting small tumors at these frequencies was influenced by skin-layer attenuation and the depth of tumors, which was not fully addressed in this study. The antenna did not offer a directional pattern.

Moreover, the pattern should be outward of the body so as not to enhance coupling between the body and the antenna. [118] designed a compact, flexible Vivaldi antenna, and an array of nine identical antennas of this type was used as a microwave breast imaging model to detect cancerous tumors in the multilayer phantom model presented in this paper. The proposed modified Vivaldi antenna had a very compact size of 25 × 20 × 0.1 mm3 and had a different geometry than conventional Vivaldi antennas. It resonated at two resonant frequencies at 4 and 9.4 GHz. While phantoms simulated the properties of human tissue, they could not fully replicate the complex biological variability of real human breasts, such as varying tissue densities, fat content, and heterogeneity in tumor types and locations. Moreover, using a single frequency range limited the resolution and accuracy of imaging, especially for detecting smaller tumors located deeper within the breast tissue. While the study demonstrated potential in a controlled experimental setting, further clinical validation on real human subjects was essential to confirm its reliability, safety, and effectiveness in diverse patient populations. A new textile-based UWB microstrip patch antenna was developed and modeled to operate within the 2–11.6 GHz frequency range, with simulations to evaluate its performance for early breast cancer detection. The antenna had dimensions of 31 × 31 mm^2 and was constructed on a denim substrate, along with a 100 % metal polyamide fabric composed of copper, silver, and nickel, ensuring wearer comfort. The antenna's return losses for the four models ranged from −36 to −18.5 dB, depending on the tumor's size and position. Flexibility tests, including bending effects, confirmed that the antenna was highly efficient and could detect tumors as small as 2 mm in diameter. The system was primarily tested using breast phantoms, which did not fully replicate the complexities of real human tissue, especially regarding different tumor types and body variations. It also lacked extensive clinical trials on real patients, critical for confirming its performance in diverse, real-world conditions. Moreover, despite their results, the defective ground structure created a directional radiation pattern, which was unacceptable since those radiations were parallel to the body, which could enhance the coupling between the body and the antenna and degrade its performance [119].

4.2. Breast cancer imaging-based monopole antennas

A breast cancer imaging-based monopole antenna is a type of antenna specifically designed for breast cancer imaging. It is typically tiny and planar, made of a metallic material printed on a substrate. The antenna is used to send and receive radio waves and is placed on the surface of the breast. The radio waves are then used to image the breast tissue. Monopole antennas have several advantages that make them suitable for breast cancer imaging. First, the antenna is relatively easy to make and use. Second, the antenna is susceptible to radio waves, allowing high-quality images to be produced. Third, the antenna is inexpensive, making it an economical option for breast cancer imaging.

However, there are also some detriments to using a breast cancer imaging monopole antenna. First, the antenna has a low gain, and the radiated wave may get a lot of reflections on the skin surface and not penetrate enough into the breast. Second, since the pattern is omnidirectional, other radio wave sources can interfere with the antenna, which can degrade the image quality. Third, some patients may find the antenna to be uncomfortable. Some of these UWB monopole antennas are presented as follows for better investigation and understanding.

Utilizing a planar UWB patch antenna array for breast tumor detection entails analyzing microwave signals' interaction with breast tissue to locate abnormal tissue, such as tumors, through signal reflections, scattering, and attenuation analysis. The antenna obtained a gain of over five dBi across the entire 3.1 GHz to 12 GHz utilizing a frequency-selective surface (FSS) layer to have an early time detection. However, this technique enhanced the antenna's profile, and the obtained gain was inadequate compared to the antenna's size and layers. Besides, the study primarily tested the antenna with breast phantoms but did not fully represent the complexities of actual human tissue. It operated in specific frequency bands, potentially limiting its effectiveness in detecting tumors at varying depths and sizes.

Furthermore, no clinical trials on real human subjects were conducted, which is essential to confirm the system's practical application [120]. A compact planar UWB antenna was developed for breast tumor imaging. The antenna's effectiveness in tumor detection was ensured through design, fabrication, and inspection. Despite its small size, it maintains stable radiation characteristics with a reasonable gain, having a size of 40 × 40 mm^2. The paper mentioned minor discrepancies between simulated and measured results, mainly due to manufacturing tolerances and connector issues. These inaccuracies affected the antenna's performance and reliability in real-world applications. It indicated potential challenges in achieving consistent, linear phase characteristics across the operating band. Therefore, it could not detect the tumor with high precision, and several clutters detected along with the cancer itself and at its lower operating frequency range hampers imaging quality and limits the likelihood of obtaining high-resolution images of breast tumors. It was due to the antenna structure and its defected ground [121]. Another UWB monopole antenna showed potential performance for imaging purposes due to its multi-wideband and miniature size. It could obtain the working bands of 2.38–2.42 GHz and 3.35–11 GHz. However, its focus on Bluetooth and general UWB applications limited its applicability for breast cancer detection, requiring a specific range of frequencies for practical imaging. Variations in breast tissue properties and the anatomy of different individuals could result in mismatch issues that could degrade the antenna's performance when used in real-world imaging scenarios, and they did not consider this factor. This antenna's frequency bands could not be ideal for the susceptible, low-signal environments required for breast cancer detection. The antenna's sensitivity to environmental factors such as body tissue and external noise was not optimized [122].

Furthermore, [123] crafted a compact UWB antenna with dual functionality, ensuring strong signal quality both on and off the body. This design includes a circular ring with six intersecting rings in the emitting part and a small elliptical ground plane. Shrinking its size, the final design (Antenna 3) achieved signal quality over 0.93 near the human arm, fostering effective in-body communication. The antenna obtained a BW of 3–12 GHz and a gain of less than five dBi. Results revealed successful detection of cancerous tumors of 6 mm and non-cancerous tumors of 3 mm. Yet, tumor responses are masked by skin reactions. Hence, tumor signals are concealed within background noise or unwanted signals from skin tissue reflections in the imaging system. However, the broad coverage of the antenna could be problematic for breast cancer since it could radiate in other directions as well. This issue was evident in the reconstructed image of the tumor, which detected more clusters and artifacts in the imaging area of the breast.

Furthermore, the antenna used an FR4 substrate, which could introduce higher dielectric losses, affecting performance in complex tissues. The evolution of its structure through multiple steps increased fabrication difficulty, and the breast tumor imaging results lacked extensive validation with realistic clinical scenarios or diverse phantom models. In addition to this design, the authors in [124] examined the impact of the fidelity factor on tumor detection in microwave imaging systems. While important, this narrow focus overlooked other critical parameters, such as efficiency, gain, or practical usability in real-world imaging setups. Besides, it did not fully address challenges like tissue interaction, resolution, and accuracy in detecting small or deep-seated tumors. The study primarily used simplified or homogeneous phantoms; the findings could not translate effectively to real-world, heterogeneous breast tissue environments.

4.3. Breast cancer imaging-based Vivaldi antennas

The Vivaldi antenna stands as a unique variant of a dipole antenna, recognized for its wide bandwidth and impressive gain. Its adaptability

to contour along the breast's shape makes it particularly suitable for breast cancer imaging applications. This flexibility enhances signal penetration and reception profoundly. The Vivaldi antenna can be pivotal in breast cancer imaging, positioned upon the breast's surface to transmit and receive radio waves. These radio waves intricately engage with the breast tissue, subsequently rebounding towards the antenna. The antenna metamorphoses these reflected radio waves into discernible electrical signals through a remarkable conversion process. These signals, in turn, pave the way for constructing a comprehensive breast image, offering valuable insights for diagnostic purposes. A spectrum of studies attests to the efficacy of Vivaldi antennas in breast cancer imaging. The advent of Vivaldi antennas heralds a propitious stride forward in breast cancer imaging technology. Their inherent strengths, from their wide bandwidth and substantial gain to their seamless adaptation to anatomical contours, confer distinct advantages over alternative imaging modalities. However, the substantiation of these promising benefits necessitates continued research, primarily through extensive clinical trials, to solidify the role of Vivaldi antennas in revolutionizing breast cancer imaging.

In [125], a compact Vivaldi antenna was designed through the fractal slot to detect strokes in the human head. The antenna was designed with an overall dimension of $65 \times 65 \times 1.6$ mm^3 using Roger RO4350B. They achieved a bandwidth of 1.44 GHz ranging from 2.35 GHz to 3.79 GHz and a maximum gain of 7.35 dBi. The antenna's narrow band was not optimal for breast microwave imaging, often requiring lower frequencies (e.g., sub-2 GHz) to achieve deeper tissue penetration and better resolution. Moreover, the antenna's unidirectional pattern could not provide the ideal coverage for breast imaging, which often benefited from omnidirectional or wide scanning coverage to accommodate varied breast shapes and sizes.

Additionally, due to their high gain, Vivaldi antennas can receive weak signals from the breast tissue, a critical capability for imaging small or deep-seated tumors. Moreover, their convenient conformal shaping to the breast enhances signal penetration and reception, which proves vital for imaging the entire breast, encompassing regions proximate to the chest wall or ribs. Presenting a promising innovation in breast cancer imaging, Vivaldi antennas showcase numerous benefits compared to alternative imaging methods. These encompass their expansive bandwidth, substantial gain, and adaptable conformal structure.

Tumorous tissue has distinct electrical properties from healthy tissue, including permittivity, conductivity, and scatter variations. Detecting these differences can identify abnormalities like breast cancer. Microwave imaging holds promise for detection, but challenges remain [126]. Achieving precise resolution and depth to find small tumors without false positives is a challenge. Clinical trials are crucial to validate sensitivity, specificity, and reliability [127]. Notably, [128] introduced two key contributions. Firstly, they proposed "complex natural responses" (CNRs) to differentiate benign and malignant tumors based on morphological appearance. CNRs successfully distinguished a 5 mm benign tumor from an averaged 5 mm malignant tumor with a speculative periphery in simulations. Secondly, they presented microwave images of both tumor types using a single Vivaldi antenna in a mono-static system. This system located tumor positions in resulting images. The antenna rotated around the breast in this proposed system, keeping a constant skin distance for signal transmission and reception. It maintains a nearly uniform amplitude for skin effect in received signals. Meanwhile, as the antenna nears the tumor during rotation, the tumor response amplitude strengthens but weakens as the antenna moves away from the tumor. Nevertheless, the antenna dimensions were large which reduces the comfortability and not many arrays of the antenna can be used so as less accuracy in detection of the tumor. Furthermore, this approach was impractical for clinical implementation as an identical tumor-free breast is not accessible.

Similarly, [129] proposed a new compact UWB cylindrical slotted Vivaldi antenna for a microwave imaging system to detect breast cancer. The hemi cylindrical slots on the edges of the Vivaldi antenna increase its electrical length and focus the energy on the aperture center. This mixture of the hemi cylindrical slots and directors improved UWB impedance matching and its gain and directivity. The time domain study of the antenna also presented good pulse handling ability with nearly flat group delay over the operating UWB band. While it offered ultra-wideband (UWB) performance, the frequency band used was not fully optimized for the varying dielectric properties of breast tissues. This could affect the depth of penetration and resolution in heterogeneous breast tissues. The paper did not extensively detail SAR (specific absorption rate) evaluation, which is crucial for ensuring the antenna's safety in sensitive applications like breast imaging. Furthermore, the paper primarily relied on simulations, which might not fully replicate the complexities of human breast tissues or environmental conditions encountered in practical imaging scenarios.

The Frequency Selective Surface (FSS) was reduced in size to enhance the gain of a compact UWB monopole antenna for microwave imaging applications. The designed hexagonal antenna was printed on an FR4 substrate with a coplanar waveguide feed and was supported by a 3×3 FSS array positioned 21.6 mm behind it. The FSS exhibited a transmission magnitude below -10 dB and a linear reflection phase across the 2.6 to 11.1 GHz frequency range. The antenna prototype demonstrated a significant gain improvement of approximately 3.5 dBi, with overall dimensions of $110 \times 140 \times 70$ mm^3. Incorporating FSS increased the design complexity, as the optimal gap between the antenna and FSS must be carefully maintained. The antenna and FSS were fabricated using an FR4 substrate with suboptimal dielectric properties (e.g., high loss tangent) at microwave frequencies. Furthermore, its sensitivity to low-contrast targets (such as early-stage breast tumors) remains unexplored. This is a critical performance metric for effective microwave imaging in medical diagnostics [130]. [131] introduced the Compact Ultra-Wideband Circularly Tapered Vivaldi Antenna with the Director (CTVAD) for microwave-based breast tumor detection. CTVAD stands out due to its compact size, enhanced impedance bandwidth, and precisely directed radiation pattern. It maintained consistent group delay, over 90 % system fidelity, and a Front-to-back ratio (FBR) value above 9 dB by utilizing directors between the antenna flares. Offering these performances, the antenna had a bidirectional radiation pattern, which affected the reconstructed images of the tumor detecting and clutter around the cancer. High-gain antennas could be more prone to interference and clutter in heterogeneous tissue environments, complicating signal interpretation. The paper emphasized simulations but lacked extensive *in-vivo* or phantom-based experiments, which were critical for verifying the performance in realistic breast tissue scenarios. Finally, the design did not discuss detailed bio-compatibility or safe integration for use near human tissues, which is crucial for medical imaging.

A high-gain microwave breast imaging system utilizing metamaterial-loaded antennas has been developed to detect breast phantoms' tumors. The proposed design is based on a modified version of a primary Vivaldi antenna. These antennas achieved a -10 dB impedance bandwidth of 8.45 GHz, from 2.0 to 10.45 GHz, and a maximum gain of 7.9 dBi within the operating band. The fidelity factor for face-to-face orientation was calculated to be approximately 92.89 %. The antennas were designed with dimensions of 46×40 mm^2. However, real-world testing with human tissue models is necessary for more accurate assessment, and it did not address the complexities of phantom simulations. Using nine antennas in a complex array setup could increase the system's overall cost and complexity. These factors might limit practical deployment in clinical settings. Achieving high-resolution imaging in real-world conditions, especially for small tumors, requires further refinement of the system's design and processing techniques [132].

It is worth noting that some designs have employed advanced techniques and structures, such as magnetic conductors and metamaterials, to enhance the performance of microwave imaging antennas. For

instance, in [133], an artificial magnetic conductor (AMC) was integrated to improve frequency selectivity at 8.6 GHz, achieving a peak realized gain of 9.9 dBi. However, combining the AMC layer requires precise alignment and physical separation from the antenna, which could be challenging in practical applications. The resonance frequency of 8.6 GHz was adequate for the system's design but was not optimal for all imaging applications. Different tumor sizes, depths, and tissue types could require a broader frequency range for more accurate imaging.

Similarly, [134] used a rectangular patch antenna with a Circular Split Ring Resonator (CSRR) in the ground plane, demonstrating promising results. Although THz radiation had high transmission properties in thick and fatty tissues, there were still limitations regarding its penetration depth, especially in more heterogeneous tissues like those found in the breast. Further research was needed to overcome depth resolution and tissue differentiationissues. While the antenna design was promising, integrating such a specialized system into a comprehensive imaging setup, such as those used for clinical breast cancer screening, required significant adjustments to accommodate portability, ease of use, and integration with existing technologies. Moreover, the breast's complex anatomy and tissue composition affected detection accuracy. Differences in density, shape, and size lead to signal propagation variations, possibly causing false results. External factors like motion, interference, and noise in the received signals also degraded system performance and reliability.

4.4. Planar patch antennas for breast cancer imaging

Planar patch antennas for breast cancer imaging are constructed from thin, flexible materials that conform to the breast's shape, making them ideal for microwave frequency applications. These antennas are easy to fabricate and integrate into imaging systems. Their flexibility allows them to adapt to various breast sizes and shapes, minimizing signal loss and improving image quality. Offering a cost-effective solution, these antennas have the potential to enhance breast cancer diagnostics and make imaging technologies more accessible globally.

Circular monopole-like patch UWB antennas showed promise for early breast cancer detection, operating widely for varied pulses. Circular patches have favorable traits like omnidirectional patterns and low cross-polarization. In [135], a UWB circular patch antenna design was developed to offer high-resolution imaging and deep tissue penetration, focusing on breast tumor detection. The antenna, which is FCC-approved for biomedical use, achieved a wide impedance bandwidth of 149 %, ranging from 2.9 to 20 GHz, excluding four notched frequencies. Despite its promising results for biomedical applications, the band notches, lower gain, and defected ground plane that resulted in bidirectional radiation limited the antenna's overall performance for imaging purposes. The primary focus of the antenna was on rejecting specific narrowband frequencies, which could be a disadvantage for applications like microwave imaging, where a broad, continuous spectrum was often needed for optimal resolution and imaging performance. While compact, the intricate structure resulted in manufacturing challenges, particularly in applications requiring precise, small-scale antenna designs. Although the antenna was designed for high gain in the passband, the gain dropped significantly in the stopbands. High and uniform gain across a wide frequency range is essential for imaging accuracy for medical applications like breast cancer detection.

MWI's role in early breast cancer detection reduces associated mortality. Imaging system accuracy is influenced by frequency range. In this context, Gupta et al. [136] developed a Defected Ground Structure (DGS) UWB antenna for breast tissue imaging. This antenna realized a broad frequency range, which enhanced the matching and performance through a modified radiator design. Moreover, the stable and linear behavior is suited to microwave imaging. While the antenna offered a broad frequency range, practical challenges remained in optimizing its bandwidth for the specific tissue properties and tumor characteristics encountered in real-world breast cancer detection scenarios. The antenna achieved moderate gain (around 4.43 dBi) and efficiency, which might not be sufficient in specific complex imaging scenarios, such as deeper or smaller tumors. The testing for breast cancer detection was based on simulations using a breast phantom model. Although these models were applicable, they could not fully capture the complexity of human tissue variations, such as varying permittivity or conductivity across different women. Moreover, real-world tests with diverse populations and tissue compositions could reveal additional challenges that were not present in the simplified model.

The semi-elliptical defected ground structure (SE-DGS) [137] was an inventive UWB antenna design. It paired a tapered microstrip line-fed antenna with a semi-elliptical DGS, boosting performance. Even when reduced, the antenna's physical size cannot always be optimal for all types of imaging setups, especially in scenarios that require a conformal or flexible integration with the body or where space is highly constrained. The antennas' particular structure added complexity to the manufacturing and integration processes, especially if mass production is considered. The actual imaging performance in real-world conditions might differ, and it was performed here. These variations, such as the effect of skin, fat, and muscle layers, influence the antenna's efficiency and accuracy. Because of this, [138] introduced a novel compact UWB antenna, the SE-DGS tapered microstrip line-fed design. This antenna created notches in specific frequency bands, suitable for Ku and K band integration. However, the antenna's gain fluctuated between 1.7 dBi and 4.82 dBi across the UWB range. This affected the consistency of imaging performance, especially in deep tissue penetration or complex breast tissue structures. The band rejection capability limited the use of the antenna in multi-band systems or in environments where precise control over all frequency bands is crucial.

It's small yet offers wide radiation bandwidth, fractional bandwidth, and varying gain across the UWB range [139]. The design enhanced wireless applications with an exceptional radiation efficiency of 96 %, consistent patterns over the UWB range, and impressive interference rejection within notched bands. Nevertheless, different breast tissues and tumor types were not considered, which could affect signal propagation. The antenna did not assume real-world clinical scenarios where multiple antennas are needed for full-body or 3D imaging. This variability could reduce the accuracy and consistency of tumor detection across diverse patients. Apart from those, its radiation was not all directed to the breast, and it showed low gain at lower bands.

The proposed antenna array operates within the 3.1 GHz to 11.6 GHz range under FCC regulations, allowing the imaging system to benefit from deep penetration and high resolution for precise image acquisition. The antenna array has a compact size of $21 \times 21 \times 12$ cm^3, enabling multiple elements to be integrated if their coupling stays within acceptable limits [140]. In addition, a compact ultra-wideband (UWB) patch antenna for microwave breast imaging (MWI) applications was presented. Slits were introduced in both the radiating element and the ground plane to enhance the antenna's bandwidth performance. The proposed design featured a size of $20 \times 19 \times 1.6$ mm^3, achieving a return loss of less than -10 dB across a 7 GHz bandwidth and a gain exceeding three dBi. A defected ground layer reduced the antenna's gain since it could not be directive but bidirectional.

Nevertheless, the antenna's dimensions were enormous, affecting the quality of the reconstructed image. Besides, the balance between deep penetration and high spatial resolution remained a technical hurdle. External factors, like body movements or surrounding materials, affected the consistency and accuracy of the microwave signals. In addition, the real-world clinical trials necessary to confirm the practical feasibility and robustness of the system were not tested.

The antenna in [141] featured square-shaped concentric rings connected to a central patch via a common feedline. This antenna array operated across a frequency range of 2–12 GHz, effectively covering the spectrum used by current microwave imaging systems. It demonstrated an average gain improvement of 4.8 dBi and an efficiency enhancement of 18 %. However, the relatively low gain negatively impacted the image

quality and accuracy, limiting its effectiveness for imaging applications. While the study's authors mitigate the mutual coupling issue using their proposed design, it remained a significant concern in real-world applications. At the scale needed for breast cancer detection, it could be costly and challenging in terms of mass production. Furthermore, the proposed antenna array required a complex setup, with multiple antennas strategically placed around the breast to collect and analyze scattered signals.

In the field of enhancing breast cancer detection through microwave imaging (MI), a newly designed miniaturized monopole patch antenna was employed for the Ultrawideband (UWB) frequency range. This compact antenna in [142] demonstrated a measured fractional bandwidth of 141 % and a gain increased from 1.4 to 6.43 dBi, indicating its potential for effective MI-based cancer detection. However, this design had limitations in real-world applications for breast cancer detection. These included potential issues with antenna-skin interaction due to its proximity to the human body and challenges in the scalability and integration of the antenna into a practical, portable imaging system. The antenna's performance in a clinical environment, especially in terms of signal penetration and resolution for early-stage tumor detection, also were not included in the investigation.

Additionally, [143] addressed the design of a compact Ultra-wideband (UWB) antenna for microwave imaging (MI), overcoming the limitations of traditional imaging methods. It developed a rectangular microstrip patch antenna on an FR4 substrate with corner etches and a truncated ground plane to achieve UWB characteristics. The fabricated antenna exhibited good radiation characteristics and a gain with closely aligned simulated and measured S11 values. While the antenna described as having a compact design, integrating such antennas effectively in a practical system for breast cancer detection still poses challenges. The truncation of the ground plane affected the antenna's radiation pattern and, in turn, its ability to effectively image tumors at various depths within the breast tissue.

A UWB antenna was designed on a low-cost FR-4 substrate, initially featuring improved impedance bandwidth. This enhancement was achieved by incorporating an additional slot and two T-shaped strips, which extended the upper-frequency range from 9 GHz to 15 GHz. Later, undesirable bands were filtered out, including the 4 GHz C-band, WiMAX bands at 3.5/5.5 GHz (3.3–3.7 GHz and 5.15–5.85 GHz), and WLAN systems at 5–6 GHz (5.15–5.35 GHz and 5.725–5.825 GHz). However, some limitations could arise from the size and complexity of implementing such antennas in practical systems. The antenna's sensitivity to tuning and variations in material properties made it challenging to optimize them across diverse patient sizes or anatomical conditions. Furthermore, while the dual-band filtering improved performance, the antenna's effectiveness in high-resolution imaging remained contingent on optimizing other parameters like antenna placement, depth of tissue penetration, and integration with imaging systems [144].

A compact broadband planar monopole microstrip patch antenna with an extended patch and a truncated ground plane is proposed for ultra-wideband applications. The antenna has optimal dimensions of 20 $\times$ 20 $\times$ 1.6 mm^3. It achieves an impedance bandwidth of 127 % (3.13–14.07 GHz), with the S11 parameter below -10 dB. The peak gain of the antenna is 3.4 dB, and its radiation efficiency is 78 %. This design supports frequency bands, including ultra-wideband, WiMAX, GHz WLAN, X-band, satellite communications, and other wireless communication services. However, the defected ground plane negatively affected the antenna's gain, and the resulting bidirectional radiation pattern made it less suitable for imaging applications.

Moreover, the radiation pattern and gain were not sufficiently tailored for precise tumor localization within the human body, especially considering the complex dielectric properties and varied tissue layers in breast tissue. While the antenna's low cost was an advantage, its performance might be impacted by the substrate's material loss, which could reduce overall efficiency for medical imaging applications [145]. It's essential to recognize that the ultra-wideband (UWB) patch antenna configuration offers several advantages, including broader bandwidth, higher peak gain, and a precise radiation pattern. These characteristics are crucial for generating images with improved range accuracy and enabling effective deep tissue penetration. However, a detailed examination of the antenna's specifications reveals that slight variations in the design can adversely affect its performance, particularly in impedance matching.

Moreover, a significant challenge in the antenna's design is the necessity to include more antennas to gather sufficient data from echoed signals, which is essential for accurate image reconstruction. Similarly, A novel ultra-wideband (UWB) multiple-input multiple-output (MIMO) Vivaldi antenna with dual band-notched features is introduced and fabricated. The design includes an optimized ground plane and two microstrip feed lines with a compact size of 26 $\times$ 26 mm^2. This design is recognized for its excellent radiation performance and substantial gain, making it highly efficient during operation. Experimental results show that the antenna achieves an impedance bandwidth ranging from 2.9 to 11.6 GHz, with two notched bands at 5.3–5.8 GHz and 7.85–8.55 GHz. The mutual coupling remains below -16 dB across the entire bandwidth. However, the lack of a full ground plane affected its efficiency compared to antennas with complete ground planes, potentially increasing radiation losses and reducing performance. Furthermore, the band notches introduced discontinuities that could negatively impact the antenna's UWB characteristics, particularly for imaging applications. While compact, integrating this antenna array into a real-world imaging system was not considered, especially concerning fitting the device comfortably around the breast for optimal imaging [146].

4.5. Metamaterial and frequency-selective surfaces

It must be mentioned that shrinking the antenna while retaining high bandwidth poses challenges. However, techniques like metamaterials, frequency-selective surfaces, and adaptable architectures offered solutions to compensate for it—notably, these methods-maintained performance across the antenna's intended frequency range. As a result, striking the right balance among size, performance, and durability is crucial for compact, flexible UWB antennas for breast cancer detection. A low-profile, easily integrable antenna was designed to cover the frequency range of 3.1 GHz to 10.6 GHz, ideal for low-power UWB communications. They used a coplanar waveguide (CPW) feeding mechanism, a castle-shaped profile, and a more comprehensive radiating element, improving impedance performance.

Additionally, a periodic structure-based, single-layer frequency selective surface (FSS) was introduced to enhance the antenna's gain by approximately five dBi while maintaining compact dimensions. However, the bidirectional radiation pattern of this antenna makes it less suitable for breast imaging applications. The antenna, mainly in a communication-focused design, had limited penetration in biological tissues, making them less effective for deeper tumors [147]–an innovative geometry of a microstrip sensor for high-resolution microwave imaging (MWI). The proposed design consisted of a microstrip patch antenna fed by a coplanar waveguide with a metamaterial (MTM) layer-based lens implemented on the back side and an artificial magnetic conductor (AMC) realized as a separate layer. Excellent electrical and field characteristics of the structure were confirmed through experimental validation. At the resonance frequency of 4.56 GHz, the realized gain reached 8.5 dBi, with 3.8 dBi gain enhancement contributed by the AMC having dimensions of 50 $\times$ 50 mm^2. This antenna showed promising outcomes for imaging of the breast and lung. However, the antenna was designed on a rigid substrate, and dimensions were large, so fewer arrays could be used, and less scattering data was obtained, which affected the reconstructed image and resulted in more clutter around the tumor. In addition, the use of AMC introduced nonlinear effects, such as harmonic products and intermodulation, which could lead to noise and potential image distortion [148].

Yet, integrating FSS reflectors into the antenna system poses

challenges due to intricate frequency interaction optimization. The design must also consider human-body interactions and minimize the effects of radiation. Maintaining accurate manufacturing and material standards is challenging, potentially impacting antenna performance. A coplanar waveguide-fed ultra-wideband (UWB) printed monopole antenna, designed with a strawberry-like artistic shape and paired with a single-layer frequency selective surface (FSS) as a metallic reflector, enhanced the antenna's gain. The antenna achieved a broad impedance bandwidth of 8.85 GHz, ranging from 3.05 to 11.9 GHz, covering the licensed UWB spectrum. The FSS exhibited a stop-band transmission coefficient below −10 dB, with a linear reflection phase across the 3.05–11.9 GHz range. With the FSS reflector, the UWB SAPM antenna demonstrated a gain improvement from 1.65 dB to 7.87 dB in the lower band and from 6.3 dB to 9.68 dB in the upper band, resulting in an overall gain enhancement of 6.22 dB. The gap between the antenna and the FSS contributed to a relatively stable gain response, maintaining values between 7.87 dB and 9.68 dB across the band. The antenna's dimensions are 61 mm × 61 mm × 10 mm. Despite its high performance and gain enhancement, the antenna's large size and 10 mm thickness created a bidirectional radiation pattern, making it less suitable for imaging applications. Furthermore, it did not explicitly address the need for a broad and uniform imaging area. A wide, consistent coverage in breast imaging is essential to ensure complete detection and accuracy in locating tumors. The material used as a substrate was cost-effective and easy to manufacture. Still, its performance in medical applications was limited due to its relatively high loss tangent (0.02) and low tolerance to biological tissue variations. The FSS structure added complexity to the system. While the FSS improved gain and bandwidth, it also required precise alignment and careful integration into imaging systems [149].

An elliptical-shaped director embedded in an exponentially tapered slot was used to increase gain at higher frequencies. A metamaterial-inspired structure improved gain around the center frequency and enhanced radiation pattern symmetry at lower frequencies. The antipodal Vivaldi antenna was successfully developed, featuring excellent impedance matching and high gain across an ultra-wideband (UWB) range. It achieved a maximum increase of 11.6 dBi and covered a 1–28 GHz bandwidth. Nevertheless, its large dimensions of 260 × 120 mm^2 made it impractical for breast imaging despite its high performance. Due to its high directivity and focused beam pattern, the antenna did not provide sufficient uniform coverage over the entire breast tissue. The inclusion of metamaterials and directors increased complexity, which could make the system harder to integrate and maintain in clinical settings [150].

A single antenna element featured a modified Y-shaped radiating patch, epsilon-negative (ENG) metamaterial, and a partial ground plane. A pie-shaped decoupling structure was added at the back of the radiating patch to enhance isolation between the array elements. Each antenna element was fabricated on a low-cost FR-4 substrate with 28 mm × 23 mm dimensions, covering the full UWB spectrum from 3.1 to 10.6 GHz. The simulated and measured MIMO performance metrics, including Channel Capacity Loss (CCL) < 0.35, Envelope Correlation Coefficient (ECC) < 0.0025, and Total Active Reflection Coefficient (TARC) < −11 dB, were within acceptable limits for both symmetric and nonsymmetric configurations. The design achieved a maximum gain of 7.5 dBi and a radiation efficiency below 70 %. However, the system used a metamaterial unit cell, adding complexity to the antenna design. This could complicate integration into practical medical devices, but more straightforward, more compact designs might be preferred. The antennas were designed for non-planar 3D configurations, which was unsuitable for the flat surfaces typically used in breast imaging, where the system's geometry needs to be uniform. The metamaterial's stopband characteristics could interfere with imaging quality if the material properties did not interact well with the human body [151].

After presenting the advantages and disadvantages of different types of UWB antennas designed for breast cancer imaging, a comparative study was conducted to analyze which type of these antennas can be more or less applicable for microwave breast imaging. As mentioned, ultra-wideband (UWB) antennas were widely used in microwave imaging systems, including breast cancer detection, because of their broad bandwidth, ability to penetrate tissues, and high spatial resolution. Various types of UWB antennas investigated in the previous sections offered unique trade-offs between parameters such as bandwidth, directivity, miniaturization, and ease of fabrication. Here is a comparative analysis of the most common UWB antenna types used for breast imaging (Table 4).

A. Vivaldi Antennas: They are tapered slot antennas known for their broad bandwidth and high directivity. They were particularly effective for UWB applications due to their traveling wave nature. Their advantages can be listed as high directivity, which helps in focusing signals and improving image resolution; Broad bandwidth (typically 3 GHz to 10 GHz), suitable for UWB breast imaging and high-resolution images of the tumors in the breast; and easy integration to the system for better imaging. However, their larger size and directional nature make them less suitable for compact imaging systems that require careful alignment for practical imaging.
B. Bow-tie antennas are simple UWB antennas known for their planar structure and wide bandwidth. Their compact design makes them suitable for wearable or portable imaging systems. They have broad bandwidth (typically 2 GHz to 12 GHz), covering UWB frequencies required for imaging, and they are Easy to fabricate and integrate into arrays. Nevertheless, they offer limited directivity, leading to lower image resolution than antennas like Vivaldi, and they have the potential for higher side-lobe levels, which can introduce artifacts in imaging.
C. Monopole antennas: They are commonly used UWB antennas due to their simple structure and omnidirectional radiation pattern. They have a Simple design and low fabrication cost, Broad bandwidth (typically 3 GHz to 10 GHz), and an Omnidirectional radiation pattern allowing 360-degree imaging coverage. However, they had lower directivity than Vivaldi and patch antennas, leading to lower spatial resolution. Besides, they can pick up more interference from surrounding tissues due to the omnidirectional pattern.
D. Planar Inverted-F Antennas (PIFA): They are famous for compact and low-profile designs and are commonly used in wearable devices. They have a compact and low-profile design, making them suitable for portable or wearable breast imaging systems. They showed reasonable bandwidth (3 GHz to 7 GHz) for UWB imaging and low electromagnetic coupling with nearby tissues, reducing image artifacts. However, they offered lower bandwidth than other UWB antennas, and their reduced directivity could lead to less accurate imaging than Vivaldi or horn antennas.
E. Patch Antennas: They are microstrip antennas commonly used in UWB applications due to their ease of integration and low cost. Their simple, low-cost fabrication makes them accessible for mass production. They can be efficiently designed to cover UWB frequencies (typically 2.5 GHz to 10 GHz). They also have a planar structure, which is easy to integrate into wearable or compact devices. Conversely, they have low bandwidth compared to other UWB antennas like Vivaldi and Bow-Tie. They also showed lower directivity, affecting imaging resolution.
F. Horn Antennas: They are widely known for their high directivity and gain and are often used in medical imaging for focused energy transmission. They showed high directivity and gain, leading to precise imaging; they also showed high spatial resolution, which is crucial for detecting small tumors, and their broad bandwidth (3 GHz to 15 GHz) is ideal for UWB applications. Yet, they have large sizes and bulky designs, making them unsuitable for portable or wearable devices and expensive to manufacture compared to simpler antennas like monopoles or patch antennas.

For more clarity, the presented literature is tabulated concerning

Table 4
Summary of trade-offs.

Antenna Type	Bandwidth (GHz)	Size	Directivity/gain	Cost	Applications
Vivaldi	3–10	Large	High	Moderate	Clinical, early detection arrays
Bow-tie	2–12	Compact	Medium	Low	Portable/wearable devices
Monopole	3–10	Compact	Low	Low	Omnidirectional imaging systems
PIFA	3–7	Very small	Low	Moderate	Wearable, low-profile applications
Patch	2.5–10	Compact	Low	Very low	Low-cost, mass-produced systems
Horn	3–15	Very large	Very high	High	High-resolution clinical settings

different parameters, as given in Table 5.

5. Clinical trials

UWB antennas have shown potential in diagnosing breast cancer, particularly in improving techniques such as microwave imaging. Here are some key findings from clinical trials and studies that support the use of UWB antennas in breast cancer diagnosis:

1. A clinical trial investigated the efficacy of a UWB microwave imaging system in detecting breast tumors in a cohort of women. The trial reported a sensitivity of approximately 90 % and a specificity of around 85 % in differentiating between malignant and benign lesions. The UWB system was able to provide detailed imaging of breast tissue, highlighting areas of concern that could correlate with tumor presence. Therefore, UWB antennas, operating over a broad frequency range, can capture more detailed information about the dielectric properties of breast tissues. Cancerous tissues often have different dielectric properties than healthy tissues, allowing for better detection through microwave imaging [152].
2. A multicenter trial utilized UWB microwave tomography to assess its ability to detect breast cancer in asymptomatic women. The findings showed that the microwave tomography system could successfully identify cancers not visible in conventional mammography, particularly in dense breast tissues. The sensitivity for cancer detection was about 92 %, with a positive predictive value of 80 %. This technique benefits from the high-resolution imaging capabilities of UWB antennas, which improve the contrast between different tissue types, enabling better identification of tumors that might otherwise be missed [153].
3. A clinical trial evaluated real-time UWB imaging in women undergoing breast cancer screening. The study demonstrated that UWB imaging could significantly reduce the time needed for breast examinations while maintaining high accuracy in detecting abnormalities. The ability of UWB antennas to operate with high temporal resolution enables real-time imaging, which can be particularly beneficial in clinical settings for quicker diagnosis and reduced patient anxiety. It was concluded that while using UWB signals allows for deeper tissue penetration, it can also lead to a loss of spatial resolution in deeper breast regions. Additionally, external factors such as patient movement or the need for precise positioning of antennas might limit the system's reliability. These issues must be addressed to improve the system's clinical applicability and accuracy [154].
4. The University of Bristol developed a radar-based UWB system that yielded clinical results for breast cancer detection. The study compared images obtained with this system against standard imaging techniques, suggesting that UWB technology could enhance early detection capabilities. However, challenges included ensuring the system's accuracy in detecting small tumors and its potential need for further refinement in clinical settings [155].

Table 5
The comparison among the presented related works.

Ref	Antenna Type	Flexible	Substrate	Size (mm)	Frequency Range (GHz)	Gain (dBi)	Wearable	Imaging	Pro/Cons
[121]	Monopole	No	Rogers RT5880	43 × 43	2.30 – 11	5.8	No	IC–DAS	A monopole is wideband and small. However, due to its omnidirectional pattern, it has high cross-pol and low gain.
[123]	Monopole	Yes	FR4	16 × 20	3 – 12	4.8	Yes	MERI	
[109]	Monopole	Yes	Kapton polyimide	30 × 30	3.9 – 19	—	Yes	DAS	
[129]	Vivaldi	No	FR4	49 × 48.5	2.77 – 10.4	3 – 9.5	No	MIS	Vivaldi has a high gain, good cross-polarization discrimination, and an end-fire directive pattern. However, it is large, usually requires a balance, and could be a wide beam in one plane (H-plane).
[131]	Vivaldi	No	FR4	48 × 45	3.1 – 10.6	8.45	No	DAS, DMAS	
[107]	Vivaldi	Yes	Rogers XT8100	30 × 20	3.1 – 10.6	—	Yes	—	
[136]	Planar	No	FR4	40 × 40	3.1 – 11.5	—	No	—	
[138]	Planar	No	FR4	15 × 18	4.1 – 29.7	1.7 – 4.82	No	MIS	
[110]	Planar	Yes	PDMS	32 × 32	2.7 – 14	—	Yes	—	
[142]	Planar	No	FR4	30 × 30	3.22 – 11.92	11.4	No	MIS	
[143]	Planar	No	FR4	12.5 × 30	3.25 – 14.63	9.68	NO	MIS	
[148]	FSS	No	FR4	16 × 12	4.4 – 10	—	No	RBMI	Metamaterials offer more versatility and exotic properties but can be complex and costly, while FSS are simpler and well-suited for frequency-based applications with less complexity.

MIS: Microwave Imaging System, **IC–DAS**: Iteratively Corrected Delay and Sum, **MERI**: Microwave radar-based imaging, **DAS**: Delay & Sum, **DMAS**: Delay Multiply & Sum, **RBMI**: Radar-based microwave imaging.

5.1. Some recent case studies

The following case studies worked on UWB antenna systems for imaging of breast cancer are presented as follows:

1. MARIA® M4 System: This UWB radar system was tested in clinical trials with breast cancer patients. The MARIA M4 system uses a 60-element UWB antenna array, operating between 4 and 8 GHz, and has shown success in tumor localization and imaging with patients in clinical settings.
2. University of Bristol's Multistatic UWB Radar: Tested on actual patients, this system utilizes a hemispherical UWB radar array for microwave imaging, showing promising results in detecting breast cancer by capturing the dielectric differences between healthy and cancerous tissues.
3. Microwave Imaging with UWB Antennas (University of Calgary): Real patient trials were conducted using a UWB microwave radar-based system. The clinical study showed effective tumor detection by analyzing the contrast in dielectric properties of breast tissues.
4. A Clinical Evaluation of UWB Imaging for Breast Cancer Detection: A study conducted using a multi-static radar UWB system demonstrated the ability to detect tumors in a hospital-based clinical setting with high accuracy. This system was used to compare patients undergoing routine mammography.
5. Experimental Trials at King's College London: UWB radar technology was used on breast cancer patients, and results were compared with standard diagnostic tools like mammography. The system demonstrated good imaging quality, particularly in detecting smaller tumors.

In addition, the clinical validation of UWB microwave imaging systems for breast cancer detection has been an area of significant research due to the limitations of traditional methods like mammography. Microwave imaging systems, such as MammoWave, utilize the differences in dielectric properties between healthy and malignant tissues to detect lesions. These systems are designed to provide non-invasive, radiation-free alternatives that can be more comfortable for patients, especially for those with dense breast tissue.

In a clinical trial using the MammoWave device, which operates in the 1–9 GHz frequency range, symptomatic and asymptomatic women were recruited to test the system's accuracy. The study measured the system's ability to detect lesions by creating conductivity-weighted images and analyzing microwave signal variations. The system achieved promising sensitivity rates and could differentiate between benign and malignant tissues. However, further work is still required to establish its widespread clinical use, particularly in improving specificity and accuracy. For instance, the sample size was insufficient for conclusive generalization to broader populations, particularly given the varying accuracy of microwave imaging based on breast density. While the study aimed to assess the tissue effects on sensitivity, the results may still be impacted by the variations in tissue composition [156].

Another study reviewing UWB sensor-based microwave breast imaging systems emphasized their potential for high sensitivity in detecting early-stage breast tumors, particularly by analyzing backscattered signals. However, achieving optimal specificity remains a challenge, as false positives can occur due to the complexity of breast tissue [157]. Thus, while the sensitivity of these systems is high, clinical validation across larger populations and in more diverse settings is needed to improve specificity and overall diagnostic accuracy, making them reliable alternatives for breast cancer detection in clinical practice.

6. Discussion and analysis

The potential of UWB antennas in microwave imaging systems, particularly breast cancer detection, has garnered much interest lately. In this article, we described how UWB antennas for breast cancer imaging work and some of the most essential requirements. The frequency range UWB technology uses is several hundred MHz to several GHz. This wide frequency range is particularly effective in detecting breast cancer because it permits high-resolution imaging and the penetration of human tissue. Time-domain and frequency-domain imaging are the two basic imaging methods utilized with UWB antennas for breast cancer detection. Impulse radar and ultra-wideband pulse reflection are time-domain methods that capture reflected or scattered short-duration pulses to pinpoint breast tissue abnormalities. On the other hand, frequency-domain methods extract the frequency response from continuous waveforms like an ultra-wideband frequency-modulated continuous wave (FMCW) for use in imaging. Several factors should be considered when designing UWB antennas for breast cancer imaging systems:

1. The FCC requires that the antennas have a sizeable fractional bandwidth of at least 20 % or a bandwidth of at least 500 MHz to cover the appropriate frequency range. Smaller, more compact UWB antennas, or arrays of closely spaced components, improve resolution and reveal finer breast tissue features, both essential for effectively recognizing tiny tumor masses. UWB antennas should have intense signal penetration and low attenuation to collect reflections from inside structures, which is especially important for identifying malignant abnormalities at varying depths in the breast.
2. Antenna designs should target high radiation efficiency and minimal losses to increase the detectability of breast abnormalities since this is directly related to the system's sensitivity.
3. Establishing appropriate isolation between the transmitting and receiving antennas is essential to reduce the possibility of mutual coupling and increase image accuracy in UWB systems where numerous antennas may be utilized to improve imaging quality.

UWB breast cancer imaging systems may use a wide variety of antennas, including planar monopole antennas, Vivaldi antennas, microstrip patch or slot antennas, dielectric resonator antennas (DRAs), and log-periodic dipole arrays (LPDAs). Microstrip patch antennas feature relatively good wideband properties, a simple design, and a small form factor, making them ideal for use in imaging arrays. Tapered slot antennas, or Vivaldi antennas, are noted for their high directivity and wideband performance. Due to their low side lobes and decreased cross-polarization, they are well-suited for high-resolution imaging applications. With the proper engineering, a small DRA may provide wideband operation, minimal dispersion, and excellent radiation efficiency. Despite the potential of UWB antennas in breast cancer imaging, there are several challenges and limitations to consider:

1. Multipath propagation and undesired reflections may negatively impact the accuracy of UWB imaging systems unless careful calibration processes are performed.
2. Reconstructing high-quality images from received UWB signals requires sophisticated signal processing and imaging methods, and further study is necessary to create efficient algorithms and computational methodologies. UWB systems must comply with power limits and interference mitigation requirements to guarantee safe and dependable operation without causing interference to other wireless devices; thus, regulatory considerations are also vital.
3. UWB technology is now undergoing clinical validation and incorporation into existing breast cancer detection techniques.

Extensive clinical trials and validation studies are required to determine its efficacy and viability for widespread use. Wideband capabilities, penetration, and the possibility of high-resolution imaging are just a few of how UWB antennas benefit microwave imaging systems used to detect breast cancer.

6.1. Future directions

Developing compact, high-performance UWB antennas, particularly for microwave imaging systems in clinical settings, holds tremendous potential. In the context of breast cancer diagnosis, UWB antennas have demonstrated their value by offering non-invasive, radiation-free, and cost-effective imaging solutions. However, further research and advancements are needed to fully realize their capabilities and expand their integration into clinical applications.

(a) Enhancing Antenna Design and Performance: The miniaturization of UWB antennas without compromising performance remains a crucial focus for future research. Achieving a balance between compact size, wide bandwidth, and high gain will enable more efficient microwave imaging systems that can be seamlessly integrated into wearable or portable devices. Researchers should explore novel materials and antenna geometries, such as metamaterials, flexible substrates, and fractal-based designs, to achieve improved radiation characteristics, higher sensitivity, and better adaptability to various breast sizes and shapes. Additionally, advanced multi-layer designs can be considered to improve penetration depth and spatial resolution while maintaining compactness.
(b) Improved Signal Processing Algorithms: UWB microwave imaging generates large volumes of data, necessitating advanced signal-processing algorithms to reconstruct high-resolution images accurately. Future work should focus on developing real-time processing techniques using machine learning (ML) and artificial intelligence (AI) to enhance the precision of breast tissue characterization. AI-driven algorithms can help differentiate between healthy and malignant tissues with greater accuracy by learning complex patterns in dielectric properties, even when data is noisy or incomplete. Moreover, improving the algorithms for signal deconvolution, time-domain analysis, and data fusion can increase microwave image resolution, contrast, and clarity.
(c) Personalized Imaging Solutions: Customizing UWB antenna arrays and imaging systems to suit different breast anatomies remains a significant challenge. As breast tissue varies in density, size, and shape, designing adaptive antenna configurations that can tailor beamforming and focus areas to individual patients is a promising research avenue. Developing software-controlled tunable antenna arrays would allow clinicians to adjust system parameters dynamically during scans, improving imaging results across a broader patient population. This personalization could improve diagnostic accuracy, reduce false positives, and enhance patient comfort.
(d) Integration of UWB Antennas in Wearable Technologies: Integrating UWB antennas into wearable devices is crucial for expanding microwave imaging beyond clinical settings. Wearable UWB-based microwave imaging systems could allow continuous breast monitoring in-home or remote environments, enabling frequent screenings and early detection in areas with limited access to healthcare. Research should focus on creating flexible, low-power UWB antennas that can maintain performance under varying conditions of movement and positioning. The design of battery-efficient systems with seamless connectivity to telemedicine platforms will also be critical to expanding access to these technologies in underserved areas.
(e) Validation and Clinical Trials: One of the biggest hurdles in integrating UWB antennas into clinical settings is extensive validation and regulatory approval. Future research should involve large-scale clinical trials that assess the efficacy, accuracy, and safety of UWB-based microwave imaging systems in real-world conditions. These trials should address specific concerns such as sensitivity to tissue variations, imaging in heterogeneous environments, and the long-term durability of the antennas and systems. Collaboration between academic institutions, healthcare providers, and regulatory bodies will be crucial to successfully translate UWB technology from lab-based research to widespread clinical adoption.
(f) Hybrid Imaging Systems: Combining UWB microwave imaging with other modalities, such as ultrasound or magnetic resonance imaging (MRI), could provide complementary information that enhances diagnostic accuracy. Future research should explore how hybrid systems can fuse data from multiple sources to give clinicians more comprehensive views of breast tissue. UWB antennas could be integrated into multi-modal imaging platforms, offering functional and anatomical data while reducing the inherent limitations of any single technology.
(g) Addressing Safety and Regulatory Concerns: Patient safety is paramount as UWB antennas move closer to clinical and consumer applications. Regulatory bodies require a thorough assessment of UWB emissions regarding tissue heating, electromagnetic interference, and long-term exposure. Research in this area should focus on creating low-power, high-performance designs that meet international safety standards. Additionally, ensuring compliance with radio spectrum regulations for medical devices will require ongoing collaboration with policymakers and standard-setting organizations.

7. Conclusion

The use of UWB antennas in microwave imaging systems for diagnosing breast cancer has much promising. These antennas provide several benefits over some of their traditional counterparts, such as a more comprehensive frequency range of operation, smaller size, and lower cost. Their ability to send and receive signals at varying frequencies improves breast cancer diagnosis and characterization by providing more precise information about breast tissue. Using UWB antennas, microwave imaging systems may capture sharp, deep-penetration pictures with excellent contrast. These antennas aid in the early identification and localization of breast cancers by taking advantage of the UWB signals' unique qualities, such as their capacity to penetrate multiple layers of tissue and discriminate between healthy and malignant tissues depending on their dielectric properties.

Breast cancer diagnostics would benefit significantly from non-invasive, radiation-free imaging methods, such as those made possible by the UWB imaging system. Screenings may be performed more often using UWB-based microwave imaging devices, sparing patients the pain and potential health hazards associated with repeated exposure to X-rays. It is also possible to incorporate UWB antennas into wearable or portable devices, making breast cancer screening outside clinical settings more accessible and comfortable. Particularly in geographically distant regions with limited access to healthcare institutions, this allows for the option of remote monitoring and early intervention.

Significant progress has been made in using UWB antennas to detect breast cancer. However, there are still obstacles to overcome. Among these obstacles include the need to resolve regulatory and safety issues, increase the resolution and sensitivity of imaging systems, and tailor antenna designs to various breast sizes and shapes. In addition, UWB-based microwave imaging systems need more testing to prove their efficacy and durability in real-world settings. UWB antennas have recently shown promise as a solution for breast cancer screening using microwave imaging systems. In addition, more studies in this area might lead to a dramatic improvement in breast cancer screening, which would have a direct impact on saving lives via earlier detection and treatment.

CRediT authorship contribution statement

Tale Saeidi: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sarmad Nozad Mahmood:** Investigation, Conceptualization. **Sahar Saleh:** Writing – review & editing, Visualization, Validation, Data curation, Conceptualization. **Nick Timmons:** Writing – review & editing, Supervision, Resources, Methodology,

Funding acquisition, Formal analysis, Data curation. **Ahmed Jamal Abdullah Al-Gburi:** Visualization, Software, Investigation. **Faroq Razzaz:** Resources, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Tale Saeidi reports article publishing charges and equipment, drugs, or supplies were provided by Atlantic Technological University. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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